CHAPTER 67 — FLIGHT CONTROLS — DIGITAL AFCS (SAR)

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FLIGHT CONTROLS — DIGITAL AFCS (SAR)

DIGITAL AFCS FLIGHT CONTROLS

67-1. DIGITAL AFCS FLIGHT CONTROLS.

67-2. DESCRIPTION.

This chapter contains information and instructions for maintenance of digital automatic flight control system with search and rescue configuration (DAFCS) (SAR) which includes the standard flight controls plus added (DAFCS) (SAR) components in the collective, cyclic, and directional control systems. Mechanical linkage systems, actuated by electrical actuators and conventional helicopter controls are used to control flight attitude, altitude and direction.

Standard flight control systems include dual flight controls, consisting of cyclic control sticks for fore and aft and lateral controls, collective pitch control sticks for vertical control and antitorque pedals for directional control. An aerodynamically actuated elevator provides stability about the longitudinal axis of the helicopter. Force trim systems consisting of a rotary trim actuator and a force gradient are connected in parallel to the cyclic and directional controls.

The cyclic stick centering indicator system consists of two proximity switches and two cyclic centering (CYC CTR) caution lights. The lights are located on the pilot and copilots instrument panel. System circuitry is activated whenever the rotor RPM warning light is illuminated. During ground operations below normal operation range, excessive cyclic inputs will illuminate the (CYC CTR) caution lights. Properly positioning the cyclic stick will extinguish the lights.

The cyclic control system incorporates four electrical (SCAS) servoactuators, two connected in series in the pitch system and two connected in series in the roll system. Four transducers, two in roll system and two in pitch system, are used to convert mechanical movement to electrical signals for input of pilot control commands to the AFCS. A force trim system is connected to both cyclic pitch and roll control system.

The antitorque directional system incorporates two electrical (SCAS) servoactuators, a force trim system, an electromechanical rotary actuator and two transducers that are used to convert mechanical movement to electrical signals for input of pilot control commands to the AFCS. The servoactuator is located forward of the directional hydraulic servoactuator. The transducers and force gradient are connected to the bellcrank located aft of pilot directional control pedals.

Forward servoactuator in each cyclic system and directional actuator are controlled by helipilot, HP-1, system. Aft actuators in each system are controlled by helipilot, HP-2 system.

The collective control system incorporates a force trim system consisting of a rotary trim actuator, force gradient, and linkage connected in parallel to the collective controls, and two transducers that are connected to the collective jackshaft.

Refer to Chapter 98 of this supplement for (DAFCS) (SAR) systems wiring diagrams.

67-3. TROUBLESHOOTING.

Refer to BHT-412-MM Chapter 29 for flight control troubleshooting.

FLIGHT CONTROLS RIGGING

67-4. RIGGING — POWER LEVER CONTROLS.

Refer to BHT-412-MM Chapter 76 for rigging power lever controls.

67-5. RIGGING — COLLECTIVE CONTROLS.

SPECIAL TOOLS REQUIRED

NUMBER	NOMENCLATURE
Commercial	Spring Scale (Range to Measure 4.0 to 4.5 lbs. — 17.79 to 20.02 N)

NOTE

All (DAFCS) (SAR) configurated helicopters will have dual controls installed. Refer to BHT-412-IPC for illustrated parts list.

1. Comply with the following general procedures when rigging collective controls:

a. When a rigging procedure requires a specific position to accomplish following steps, manually hold pilot collective in position and secure with collective friction adjustment.

b. Tolerance for rigging dimensions is ± 0.03 in. (0.76 mm).

c. All adjustable control tubes shall have a maximum of 1.00 in. (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, the threads must be engaged sufficiently to cover holes.

d. Apply corrosion preventive compound (C-101) to threads of all adjustable end bearings and clevises after adjustment.

e. Torque control jamnuts 80 to 100 in. lbs. (9.0 to 11.3 Nm) after adjustment unless other torque is specified.

f. Control tubes must be free to rotate several degrees about longitudinal axis. Loosen jamnuts, adjust end fittings, and torque jamnuts if necessary.

2. Disconnect the following control tubes:

a. Disconnect extension tube (2, figure 67-1) from collective lever (1).

b. Disconnect control tube (4) from collective hydraulic cylinder (3).

c. Disconnect control tube (7) from lever (8).

d. Disconnect force gradient (9) from lever (8).

e. Disconnect link (11) from rotary actuator (12).

3. If available, connect hydraulic power cart to helicopter.

4. Position pilot collective stick (17) against ‘up‘ stop, hold in position and tighten friction adjuster (16).

5. Position top edge of bellcrank (6) 0.52 in. (13.21 mm) above top of support (5) as shown on detail C.

6. Adjust tube (7) to lever (8). Tighten jamnut on tube (7) clevis. Install tube (7) on lever (8).

7. Align collective hydraulic cylinder (3) using same procedures as flight control cylinders. Refer to BHT-412-MM .

8. Ensure pilot collective stick (17) is positioned as described in step (4).

9. Move actuator valve lever (19) up. Manually pull extension tube (2) up to bottom

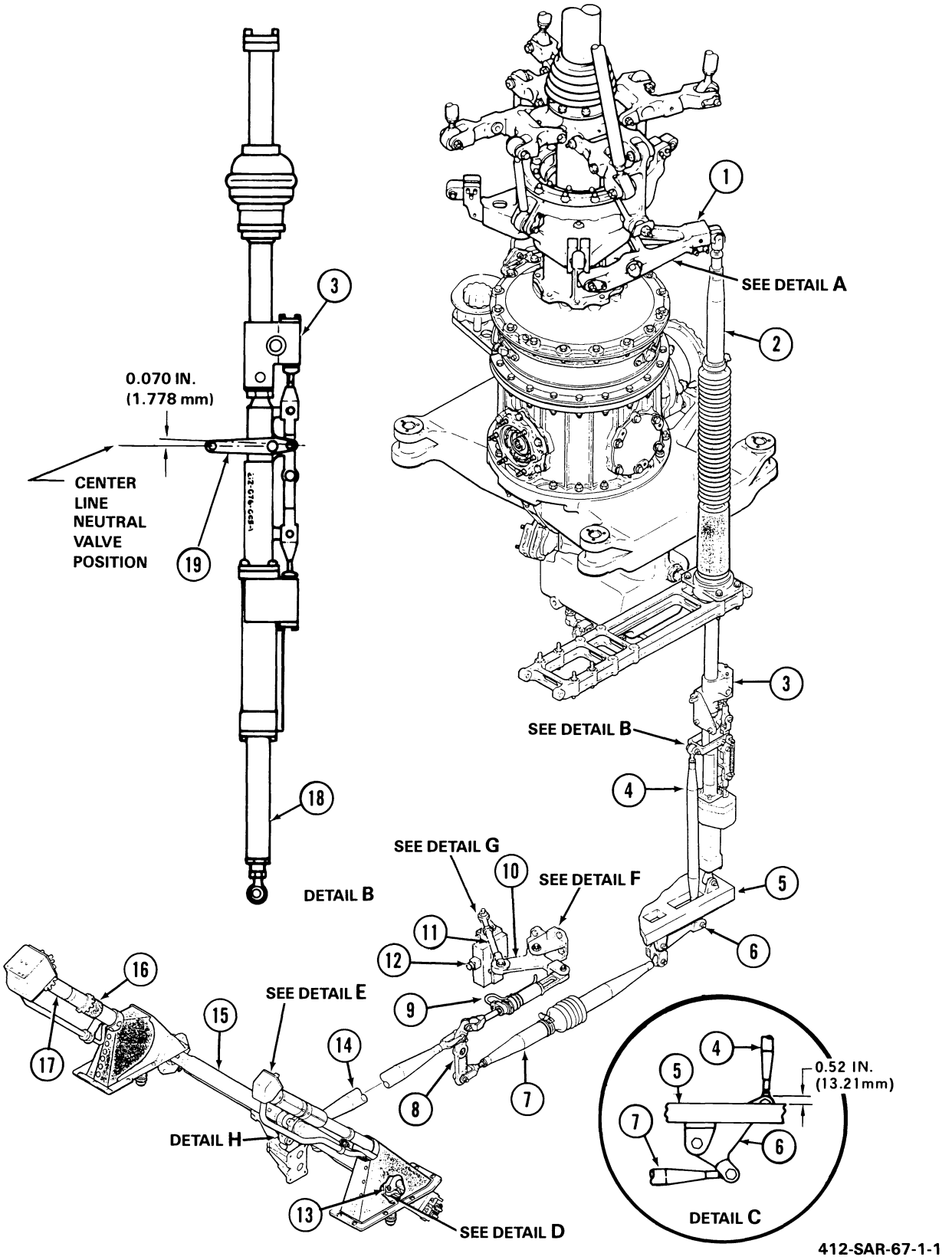
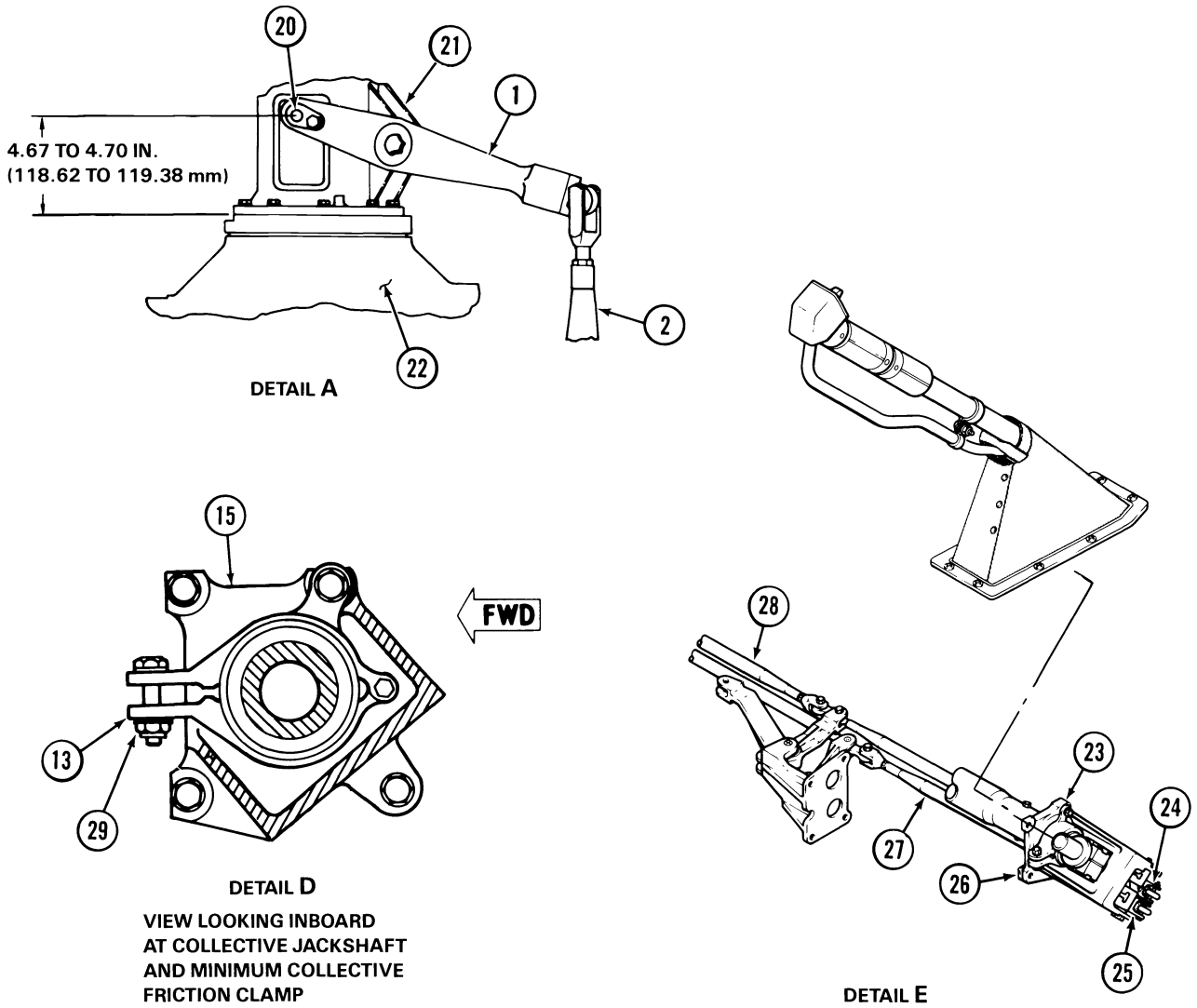


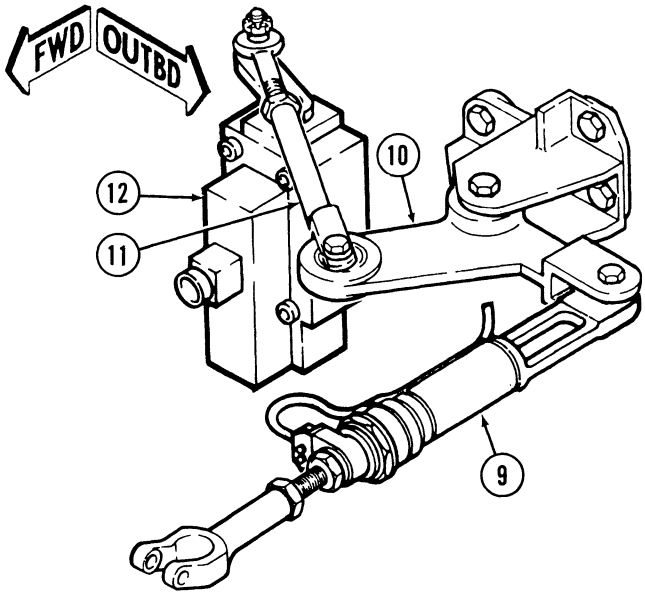
Figure 67-1. Collective controls rigging (Sheet 1 of 3)



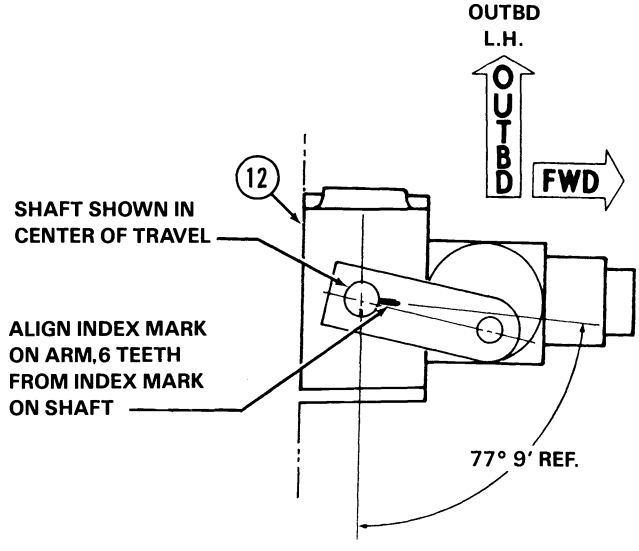
1. Collective lever
2. Extension tube
3. Collective hydraulic cylinder
4. Control tube
5. Support
6. Bellcrank
7. Control tube
8. Lever
9. Force gradient
10. Bellcrank
11. Link
12. Rotary actuator
13. Minimum collective friction clamp
14. Control tube
15. Collective jackshaft
16. Pilot collective friction adjuster
17. Pilot collective stick
18. Piston
19. Actuator valve lever
20. Pin
21. Swashplate support
22. Transmission
23. Elbow support
24. Gear sector
25. Gear sector
26. Shim
27. Control tube
28. Control tube
29. Nut
30. Collective transducers
31. Lever

412-SAR-67-1-2

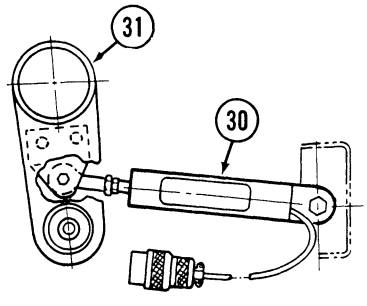
Figure 67-1. Collective controls rigging (Sheet 2)



DETAIL F
VIEW LOOKING DOWN



DETAIL G
VIEW LOOKING DOWN



DETAIL H

412-SAR-67-1-3

Figure 67-1. Collective controls rigging (Sheet 3)

collective cylinder (3) in the UP position. Then set actuator valve lever (19) 0.070 in. (1.778 mm) above center line neutral valve position as shown on detail B. Hold this position.

10. Adjust tube (4) to fit actuator valve lever (19) then shorten 3 turns of rod end bearing. Tighten jamnut against rod end bearing on tube (4). Install tube (4) on actuator valve lever (19).

11. Position collective lever (1) 4.67 to 4.70 in. (118.61 to 119.38 mm) above base of swashplate support (21) to center of pin (20) as shown on detail A.

12. Apply hydraulic power from hydraulic cart to helicopter.

NOTE

If hydraulic power cart is not available refer to Note preceeding step 16.

13. Position lever (1) as measured in step 11. Position pilot collective stick (17) against low stop. Tighten friction adjuster (16).

14. Adjust extension tube (2) to fit collective lever (1) torque jamnut 1500 to 1800 in. lbs. (168.0 to 201.6 Nm) on extension tube (2), and lockwire (C-405) . Apply adhesive (C-308) around nut, lockwire, and threads of clevis.

15. Install extension tube (2) to collective lever (1).

NOTE

Alternate method of adjusting extension tube (2) without pressurizing hydraulic system is outlined in steps 16 through 20.

16. Position collective lever (1) and pilot collective stick (17) as outlined in step 13.

17. Exert sufficient downward pressure on hydraulic cylinder (3) to hold actuator valve lever (19) at top of travel as shown on detail B.

CAUTION

COLLECTIVE CONTROLS BELOW COLLECTIVE HYDRAULIC CYLINDER (3) MUST NOT MOVE DURING ACCOMPLISHING THE FOLLOWING STEPS.

18. Adjust extension tube (2) to fit collective lever (1), then shorten by one complete turn of upper clevis in tube (2).

19. Torque jamnut 1500 to 1800 in. lbs. (168.0 to 201.6 Nm) on extension tube (2) and lockwire (C-405) . Apply adhesive (C-308) around nut, lockwire, and threads of clevis.

20. Install extension tube (2) to collective lever (1).

21. Rig 4-axis collective trim system as follows:

a. Position pilots collective stick (17) against its down stop.

b. Connect force gradient (9) to lever (8).

c. Position output arm of rotary actuator (12) as shown in (detail G).

d. Position and hold output arm of rotary actuator ((12) against its outboard left hand stop.

e. Adjust link (11) to fit between the bellcrank (10) and rotary actuator (12) positioned per step d., then extend the rod end bearing two and one-half complete turns and secure to output arm of the rotary actuator (12).

22. Perform collective transducer adjustment as follows:

NOTE

Dual element control motion transducers (412-074-101-103) are aligned and positioned using calibration tool (412-274-001-101).

NOTE

Refer to cyclic control motion transducer adjustment paragraph 67-6., 2., a., using calibration tool for electrical alignment of collective transducers

NOTE

This calibration tool (412-274-001-101) is used to electrically align the control motion transducers installed on the helicopter to be used as control position sensors for the flight data recorder. This tool will insure that the control motion transducers are aligned within ± 0.5 percent of the prescribed position. Fore and aft position is set at 50 percent. Lateral position is set at 50 percent. Collective position is set at 12 percent. Pedal position is set at 12 percent.

a. Attach collective transducers (30) to structure (see detail H).

b. Move collective transducers (30) through their full travel.

c. Position pilot collective stick (17) against the up stick stop, with collective transducers extended but not bottomed. Adjust rod ends on collective transducers (30) to align with hole in lever (31).

d. Position pilots collective stick (17) against the down stick stop, with the collective transducers (30) retracted but not bottomed, connect collective transducers (30) to lever (31). Overtravel of collective transducers (30) should be approximately equal.

23. Adjust collective minimum friction adjustment as follows:

a. Apply hydraulic power from hydraulic cart to helicopter.

b. With hydraulic test cart operating, adjust minimum collective friction clamp (13, detail D) to obtain 4.0 to 4.5 lbs. (17.79 to 20.02 N) up-collective stick force. Record this reading by applying a spring scale with a range that will measure 4.0 to 4.5 lbs. (17.79 to 20.02 N) up-collective stick force at center of upper throttle grip within ± 0.50 in. (12.7 mm).

c. Tighten or loosen nut (29) to adjust minimum collective friction clamp (13) during measure of sliding force as shown on detail D.

d. After collective friction is adjusted within limits, move collective controls through full throw and check for binding and/or obstruction of control movement.

e. Turn off hydraulic power cart and disconnect from helicopter.

24. Ensure that all collective controls bolts are installed and cotter pins and lockwire (C-405) installed, where applicable.

25. Check throttle controls for binding or obstruction as follows:

a. Disconnect left and right adjustable throttle control tubes (27) and (28) from each bellcrank.

b. Check that mark tooth of flexshaft is engaged with 10th tooth space of mating gear sectors (24) and (25) on both pilot and copilot collective elbow.

c. Roll both engine throttle grips on pilot and copilot collective sticks to full open position against stops.

NOTE

If binding or obstruction occurs in copilot collective stick throttle grips when rotated, adjust shims (26) between airframe structure and elbow support to minimize lateral deflection of elbow assembly flex shafts.

d. Adjust control tube (27) and (28) to fit. Torque jamnuts and install control tubes (27) and (28).

e. Cycle both throttle grips through full travel and check to see that each gear sector on both pilot and copilot elbow support is within no less than one tooth of rolling off its mating pinion.

67-6. RIGGING — CYCLIC CONTROLS.

Rigging procedures for the cyclic digital automatic flight control system with the search and rescue configuration (DAFCS SAR) are same as AFCS rigging. Refer to BHT-412-MM .

SPECIAL TOOLS REQUIRED

NUMBER	NOMENCLATURE
412-274-001-101	Calibration tool (pot calibrator)

NOTE

For maintenance procedures of cyclic dual controls refer to BHT-412-SI-13 and BHT-412-IPC for illustrated parts list.

1. Comply with the following general procedures when rigging cyclic controls (figure 67-2).

a. When a rigging procedure requires a specific position to accomplish following steps, manually hold controls in position and secure with pilot and copilot friction adjustment.

b. Tolerance for rigging dimensions is ± 0.03 in. (0.76 mm).

c. Apply corrosion preventive compound (C-101) to threads of all adjustable rod end bearings and clevises after adjustment.

d. Rod end bearings must be free to rotate several degrees about longitudinal axis. Loosen jamnuts, adjust end fittings, and torque jamnuts if necessary.

2. Perform fore and aft cyclic control motion transducer (34, figure 67-2 , detail A) adjustment as follows:

a. Position pilot cyclic stick (33) perpendicular to the deck within 1/2 degree longitudinally and laterally. Loosen jam nut (36) on fore and aft transducer (34) so the transducer shaft (37) can rotate freely.

NOTE

Dual element control motion transducers (412-074-101-103) are aligned and positioned using calibration tool (412-274-001-101).

NOTE

This calibration tool (412-274-001-101) is used to electrically align the control motion transducers installed on the helicopter to be used as control position sensors for the flight data recorder. This tool will insure that the control motion transducers are aligned within ± 0.5 percent of the prescribed position. Fore and aft position is set at 50 percent. Lateral position is set at 50 percent. Collective position is set at 12 percent. Pedal position is set at 12 percent.

b. Using calibration tool (38) (412-274-001-101) (pot calibrator) turn calibrator power switch (39) to the O (off) position. Connect calibrator box connector (40), to one of the fore and aft cyclic transducers (34) electrical connector (41). Turn selector knob (42) to fore and aft F/A position.

c. Turn calibrator power switch (39) to I (on) position.

NOTE

Any time the LO BAT lamp comes on, remove lid from calibrator box and replace batteries, use (9v dc alkaline batteries). Low battery is set to illuminate at 12 ± 1 volt.

d. If a RED lamp comes on, the transducer shaft (37) shall be adjusted in the direction indicated. An 'extend' indication means extend the transducer shaft (37) by screwing it into rod end bearing (43) (clockwise looking at rod end). A 'retract' indication means retract the transducer shaft (37) by unscrewing it from the rod end bearing (43) (counterclockwise looking at the rod end). Adjust transducer shaft (37) until GREEN IN RANGE lamp comes on.

e. Tighten jam nut (36) on transducer shaft (37) and verify that GREEN IN RANGE lamp is on and RED lamps are off.

f. Move calibrator power switch (39) to O (off) position. Disconnect calibrator box connector (40) from transducer electrical connector (41).

g. Connect transducer electrical connector (41) to electrical connector on the bulkhead.

h. Perform calibration adjustment on remaining fore and aft cyclic transducer (34), using steps a. through g. above.

3. Perform lateral cyclic control motion transducer (29, figure 67 detail B) adjustment as follows:

a. Position pilot cyclic stick (33) perpendicular to the deck within 1/2 degree longitudinally and laterally. Loosen jam nut (36) on lateral transducer (29) so that transducer shaft (37) can rotate freely.

b. Using calibration tool (38) (412-274-001-101) (pot calibrator) turn calibrator power switch (39) to O (off) position. Connect calibrator box connector (40), to one of the lateral cyclic transducers (29) electrical connector (41). Turn selector knob (42) to lateral LAT position.

c. Perform calibration adjustment on lateral cyclic transducer (29), by repeating steps 2. c. through 2. g. above.

d. Perform calibration adjustment on remaining lateral cyclic transducer (29), by repeating steps 2. c. through 2. g. above.

67-7. RIGGING — ANTITORQUE CONTROLS.

SPECIAL TOOLS REQUIRED

NUMBER	NOMENCLATURE
T103206	Tail rotor rigging tool
Commercial	Spring Scale (Range to measure 4 to 5 lbs. — 17.80 to 22.24 N)
412-201-700-101	Tool assembly
412-201-701-101	Tool assembly

NOTE

Rigging procedures contained herein, pertain to helicopters with (DAFCS) (SAR) with dual controls installed in basic controls systems.

NOTE

For maintenance procedures of dual controls refer to BHT-412-SI-13 and BHT-412-IPC for illustrated parts list.

1. Comply with the following general procedures when rigging antitorque controls:

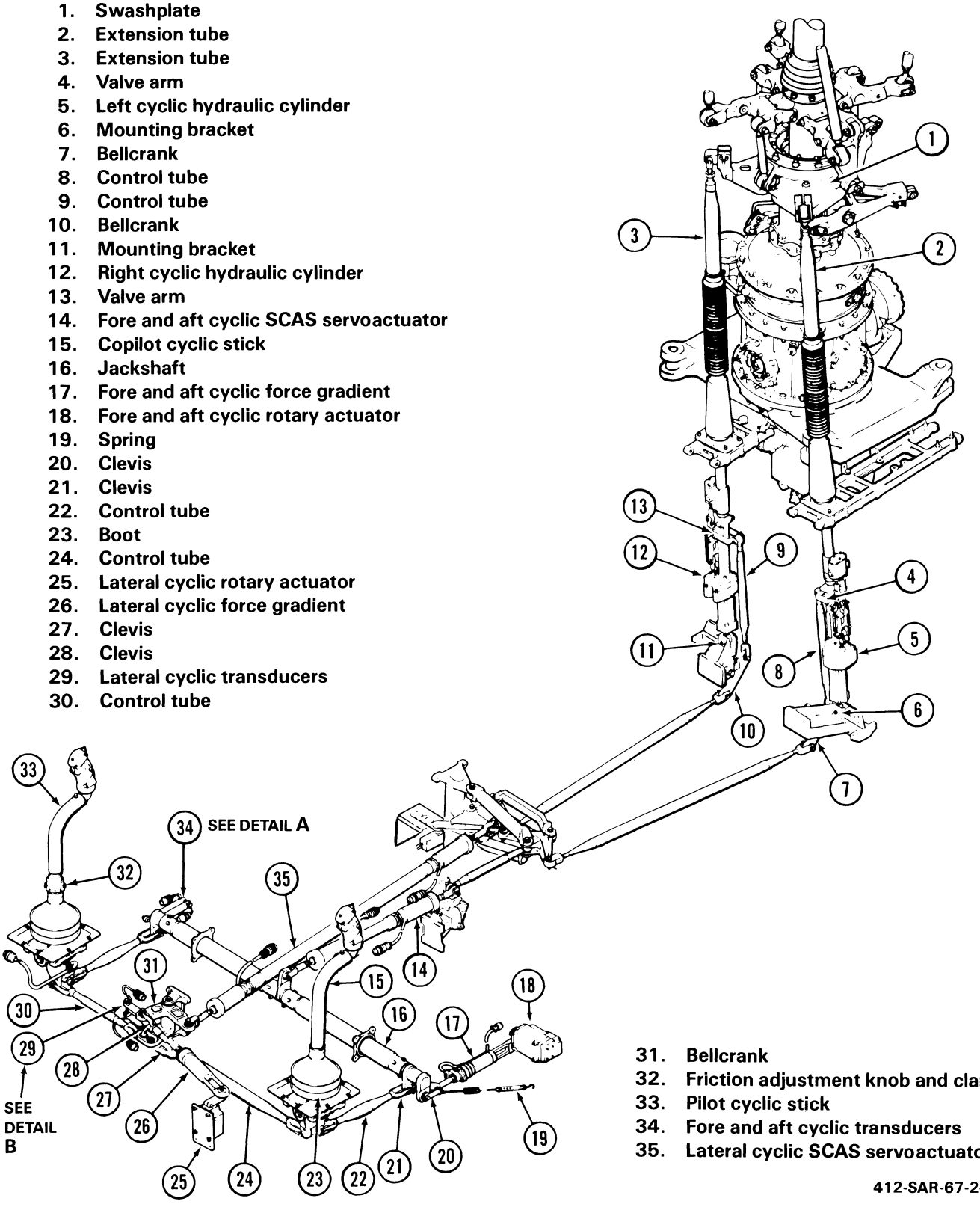
a. When a rigging procedure requires a specific position to accomplish following steps, manually hold controls in position or secure controls in position with pilot friction adjustment.

b. Tolerance for rigging dimensions is ± 0.03 in. (0.76 mm).

c. All adjustable control tubes shall have a maximum of 1.00 in. (25.4 mm) of exposed threads after adjustment. If control tubes have inspection holes, the threads must be engaged sufficiently to cover holes.

d. Apply corrosion preventive compound (C-101) to threads of all adjustable end bearings and clevises after adjustment.

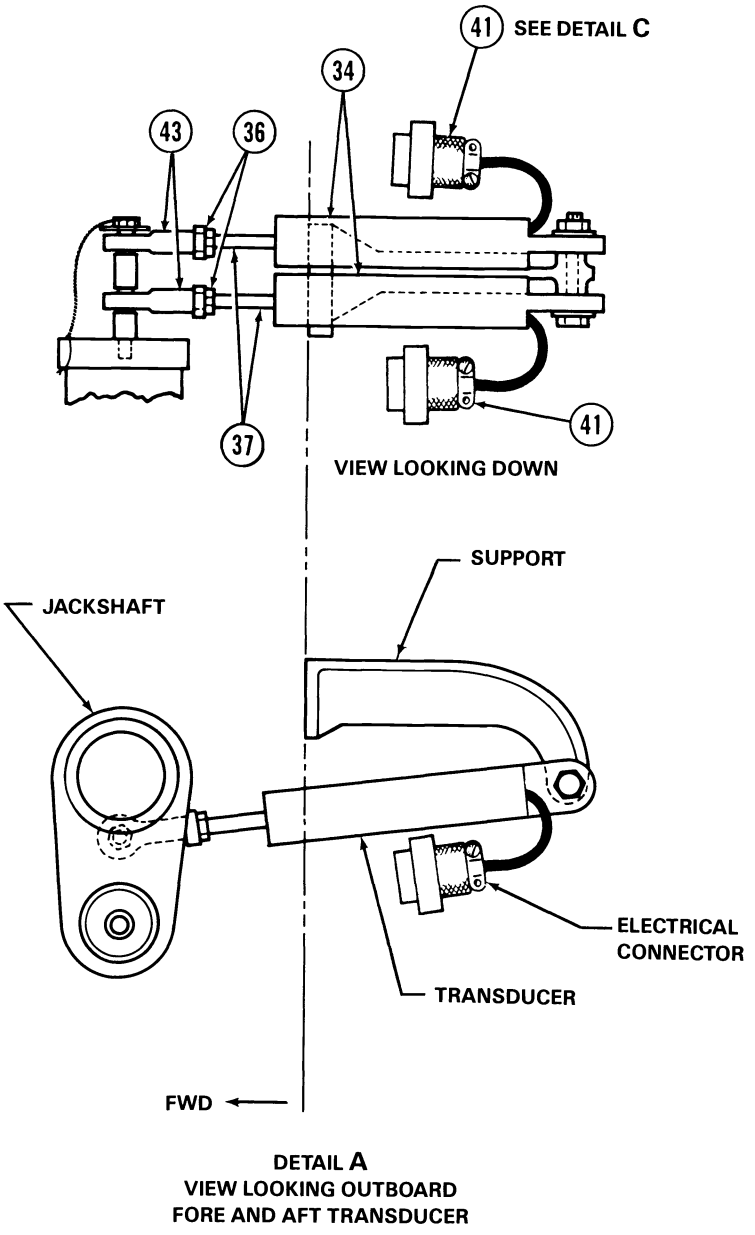
1. Swashplate
2. Extension tube
3. Extension tube
4. Valve arm
5. Left cyclic hydraulic cylinder
6. Mounting bracket
7. Bellcrank
8. Control tube
9. Control tube
10. Bellcrank
11. Mounting bracket
12. Right cyclic hydraulic cylinder
13. Valve arm
14. Fore and aft cyclic SCAS servoactuator
15. Copilot cyclic stick
16. Jackshaft
17. Fore and aft cyclic force gradient
18. Fore and aft cyclic rotary actuator
19. Spring
20. Clevis
21. Clevis
22. Control tube
23. Boot
24. Control tube
25. Lateral cyclic rotary actuator
26. Lateral cyclic force gradient
27. Clevis
28. Clevis
29. Lateral cyclic transducers
30. Control tube



31. Bellcrank
32. Friction adjustment knob and clamp
33. Pilot cyclic stick
34. Fore and aft cyclic transducers
35. Lateral cyclic SCAS servoactuator

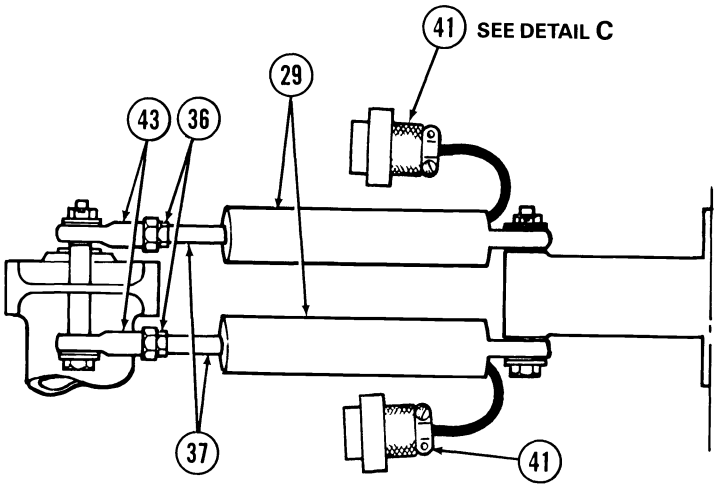
412-SAR-67-2-1

Figure 67-2. Cyclic controls rigging (Sheet 1 of 4)

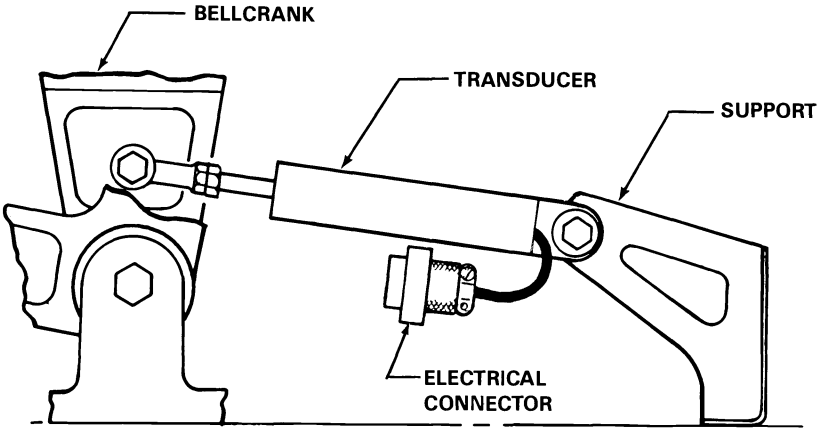


412-SAR-67-2-2

Figure 67-2. Cyclic controls rigging (Sheet 2)



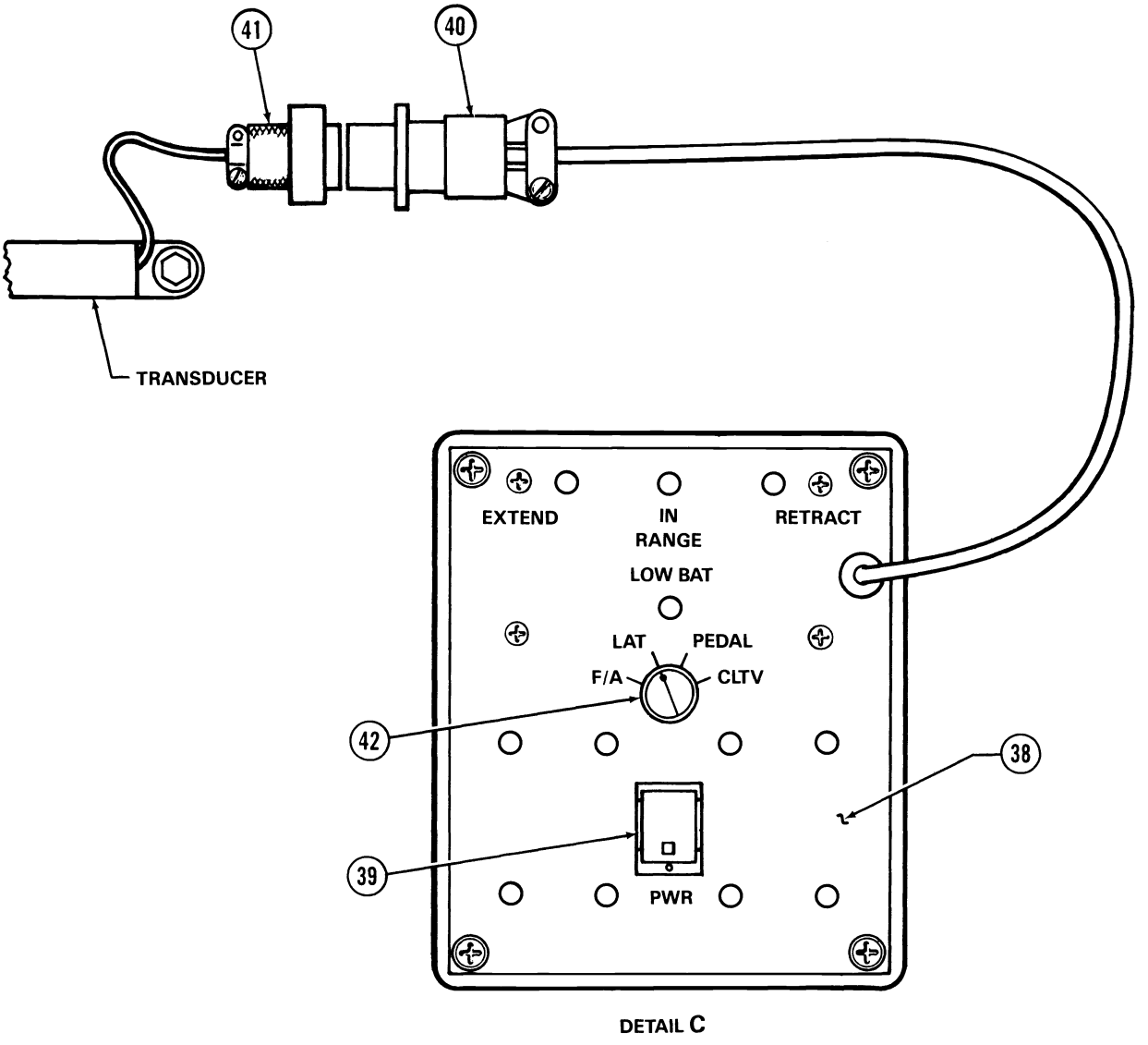
VIEW LOOKING FORWARD
LATERAL CYCLIC TRANSDUCER



DETAIL B
VIEW LOOKING DOWN LATERAL
CYCLIC TRANSDUCER

412-SAR-67-2-3

Figure 67-2. Cyclic controls rigging (Sheet 3)



DETAIL C

36. Jamnut
37. Transducer
38. Calibration tool (412-274-001-101)
39. Calibrator power switch
40. Calibrator box connector
41. Electrical connector
42. Selector knob
43. Rod end bearing

412-SAR-67-2-4

Figure 67-2. Cyclic controls rigging (Sheet 4)

e. Torque control jamnuts 80 to 100 in. lbs. (9.0 to 11.3 Nm) after adjustment unless other torque is specified.

f. Control tubes must be free to rotate several degrees about longitudinal axis. Loosen jamnuts, adjust end fittings, and torque jamnuts if necessary.

2. Disconnect the following control tubes and force gradient.

a. Disconnect force gradient (14, figure 67-3) and transducers (20) from pilot pedal adjuster (21).

b. Disconnect control tube (16) from copilot pedal adjuster (17).

c. Disconnect control tube (24) from pilot pedal adjuster (21).

d. Disconnect control tube (26) from bellcrank (25).

e. Disconnect link (4) from lever (3).

3. Connect hydraulic cart to helicopter.

4. Verify length of servoactuator (10), 62.865 to 62.915 in. (1596.77 to 1598.04 mm).

5. Apply hydraulic power from hydraulic cart to helicopter.

6. Position both pilot and copilot left directional pedals (18 and 22) against extreme forward stop. Adjust control tube (16) to fit, torque jam nut and install control tube (16).

7. Connect control tube (24) to pilot pedal adjuster (21).

8. Retract two stop bolts on bellcrank (9, detail A).

9. Install tool assembly (27, Detail B) (412-201-701-101) on pilot pedals (22 and 23).

10. Remove bolt (28) which secures servoactuator (10) to bellcrank (9), and install tool assembly (29) (412-201-700-101) and indexing pin (30, detail A).

11. Remove bolt (31) which secures bellcrank (9) to support (8, detail A).

12. Retract both antirotation screws (32) in tool assembly (29, detail A) (412-201-700-101).

13. Position tool assembly (29) (412-201-700-101) against the inboard surface of support (8). Install bolt (NAS6604-30D) through tool assembly (29) (412-201-700-101), support (8), and bellcrank (9) as shown in detail A, but do not completely tighten at this time. Rotate tool assembly (29) body until the indexing pin (30) (installed in step 10) engages the center slot of the tool assembly (29).

14. Adjust both tool assembly (29) antirotation screws (32) to make contact with the support (8) as shown in detail A. As the antirotation screws (32) are tightened, ensure indexing pin (30) is free to disengage and re-engage the center slot in the tool assembly (29) body, then tighten bolt (NAS6604-30D).

15. Remove tool assembly (27) (412-201-701-101) from the pilots left and right pedals (22 and 23, detail C).

16. Retract indexing pin (30) until it clears tool assembly (29) (412-201-700-101) body.

17. Slowly move pilots left pedal (22) forward until indexing pin (30) engages the forward slot in tool assembly (29) (412-201-700-101) body. Adjust lower stop bolt in bellcrank (9) to the left pedal stop within 0.000 to 0.002 in. (0.000 to 0.0508 mm) as shown in detail A.

18. Retract indexing pin (30) until it clears tool assembly (29) (412-201-700-101) body.

19. Slowly move pilots right pedal (23) forward until indexing pin (30) engages the aft slot in tool assembly (29) (412-201-700-101) body. Adjust upper stop bolt in bellcrank (9) to the right pedal stop within 0.000 to 0.002 in. (0.000 to 0.0508 mm) as shown in detail A.

20. Remove tool assembly (29) (412-201-700-101) and install bolt (28) and bolt (31) that were removed in steps 10. and 11.

21. Position and hold pilots left pedal (22) forward against stop.

22. Check two pitch links (2, detail E) for 6.180 in. ± 0.010 in. length (156.97 mm ± 0.254 mm) between bolt hole centers.

23. Remove bolt (34, detail E) retaining pitch link to crosshead (1).

24. Install tail rotor rigging tool (T103206) (35, detail E) to crosshead (1) and against trunnion (37). Retain with bolt (34).

25. Adjust link (4, detail F) to obtain 0.58 in. (14.73 mm) clearance with bellcrank (6). Torque jamnut on link (4), and install link (4) on bellcrank (6) and lever (3).

26. With pilots left pedal (22) positioned (per paragraph 21.), and link (4) adjusted (per paragraph 25.), adjust tube (26) to fit to bellcrank (25). Torque jamnut and install control tube (26).

27. Remove tail rotor rigging tool (T103206) (35, detail E) and install pitch link (2) to crosshead (1).

28. Rig antitorque trim system as follows:

a. Install tool assembly (27) (412-201-701-101) on pilot left and right pedals (22 and 23, detail B).

b. Position output arm (33) on rotary actuator (15, detail C).

c. Place output arm (33) of rotary actuator (15) at center of travel, adjust force gradient (14) to fit to the pilot pedal adjuster (21). Torque jam nut and install force gradient (14) on pilot pedal adjuster (21).

d. Remove tool assembly (27) (412-201-701-101) from pilot left and right pedals (22 and 23) as shown in detail B.

e. Check clearance between force gradient (14) and structure at extreme position of the output arm (33). Reindex output arm (33) relative to shaft by one seration, if required for clearance, then repeat steps b. through d.

29. Ensure that all antitorque controls bolts are installed and cotter pins, adhesive (C-308), and lockwire (C-405), is installed where applicable.

30. Perform antitorque control motion transducer (20, figure 67-3 detail D) adjustment as follows:

NOTE

Dual element control motion transducers (412-074-101-103) are aligned and positioned using calibration tool (412-274-001-101).

NOTE

Refer to cyclic control motion transducer adjustment paragraph 67-6. 2. a., using calibration tool for electrical alignment of antitorque transducers

NOTE

This calibration tool (412-274-001-101) is used to electrically align the control motion transducers installed on the helicopter to be used as control position sensors for the flight data recorder. This tool will insure that the control motion transducers are aligned within ± 0.5 percent of the prescribed position. Fore and aft position is set at 50 percent. Lateral position is set at 50 percent. Collective position is set at 12 percent. Pedal position is set at 12 percent.

31. Adjust antitorque minimum friction as follows:

a. Disconnect force gradient (14) from pedal adjuster (21)

b. Disconnect control tube (12) from walking beam (13).

c. Attach spring scale to one pilot antitorque pedal. Measure force required to move pedal through neutral. Acceptable force range is 4.25 to 4.75 lbs. (18.91 to 21.13 N). Adjust screw (39, detail G) on clamp (38) to obtain friction within acceptable range.

d. Install control tube (12) on walking beam (13).

e. Install force gradient (14) on pedal adjuster (21).

32. Move antitorque controls through full travel and check for binding and/or obstruction of control movement.

33. Ensure that all antitorque controls bolts are installed and cotter pins, adhesive (C-308) , lockwire (C-405) , installed where applicable.

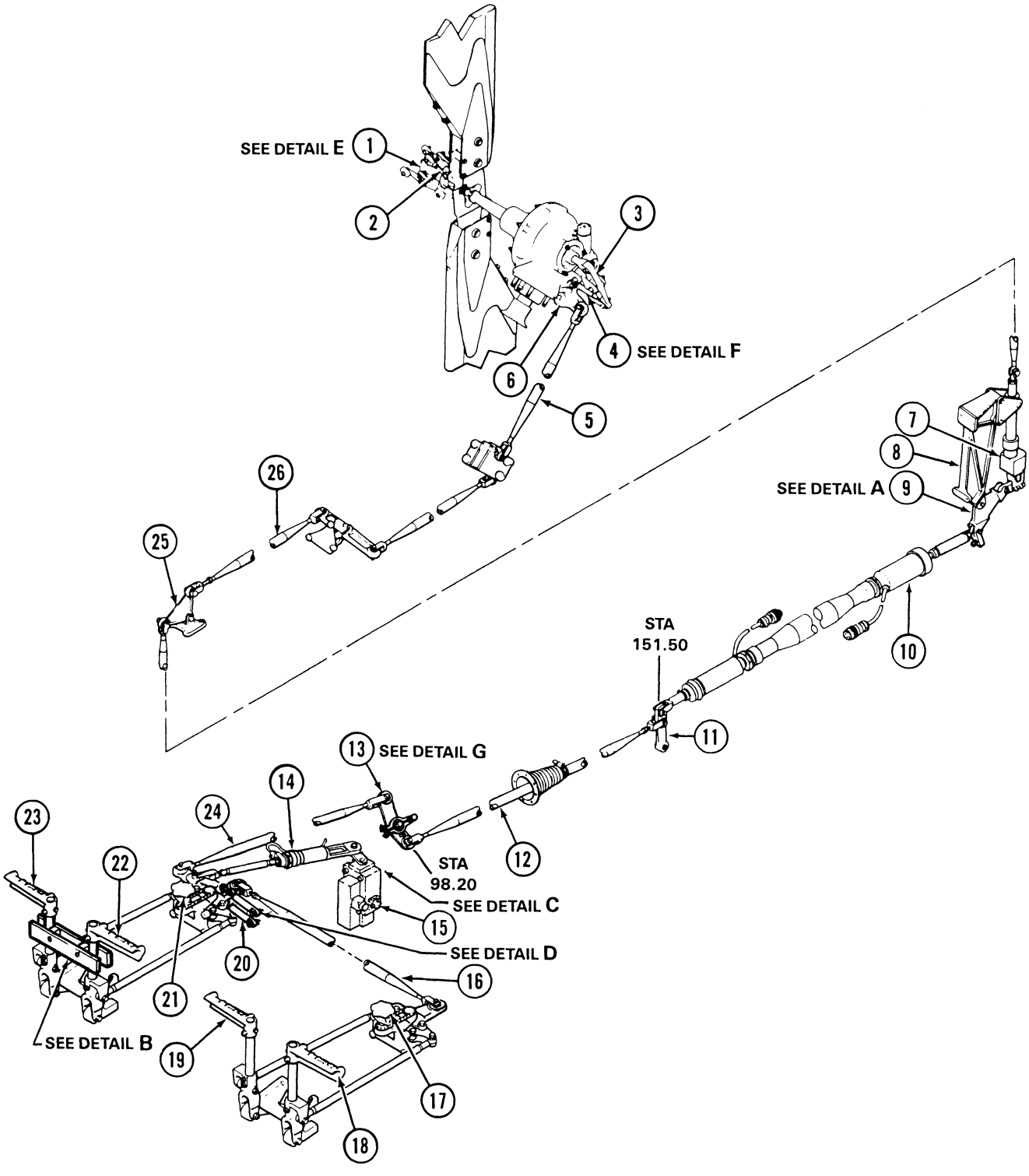


Figure 67-3. Antitorque controls rigging (Sheet 1 of 4)

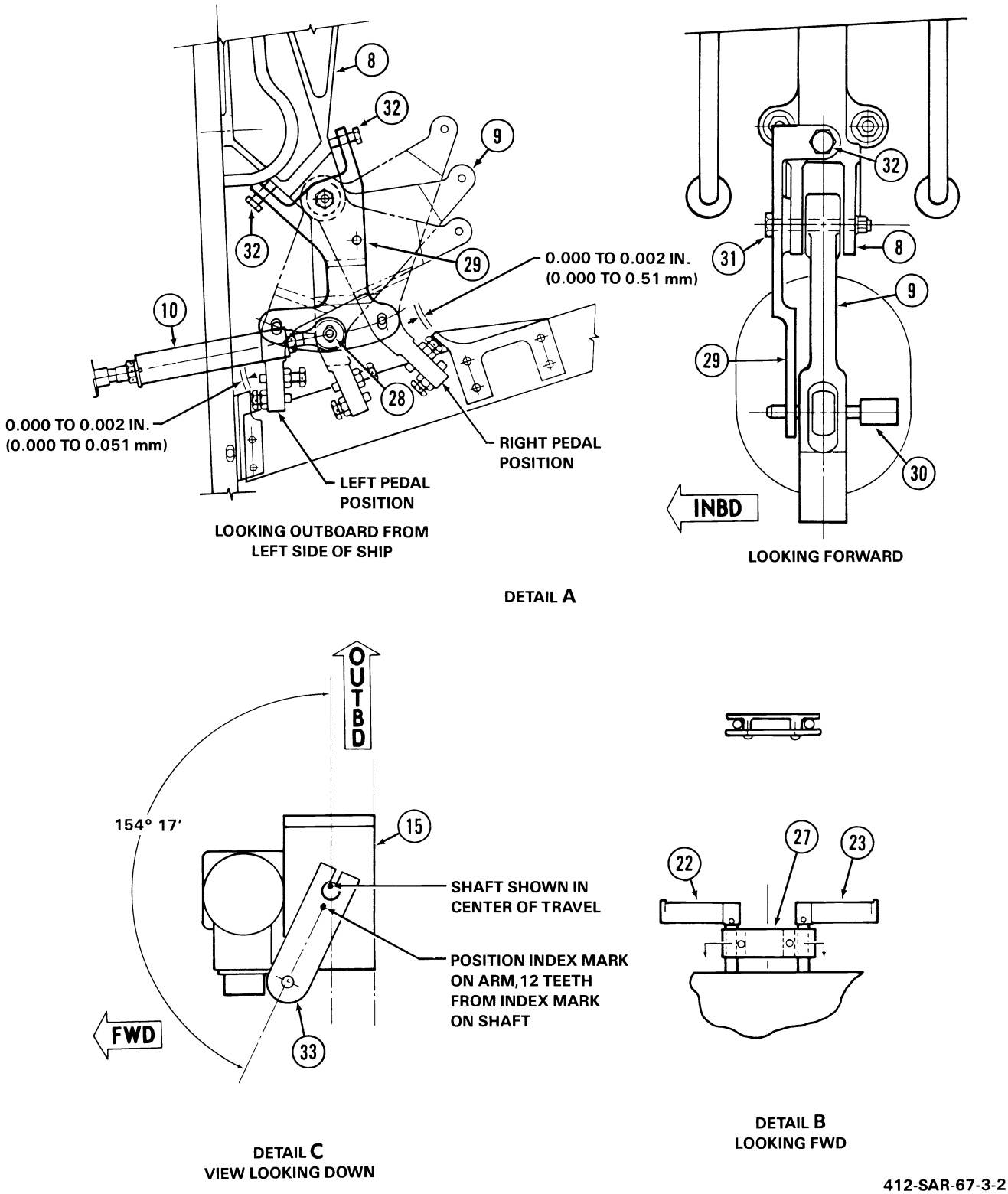
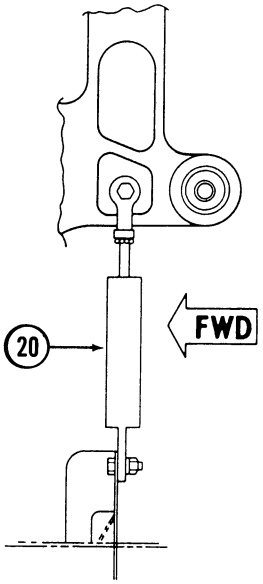
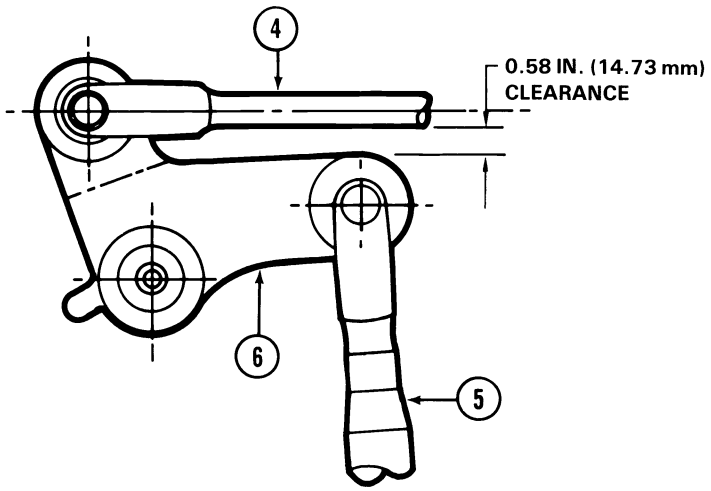


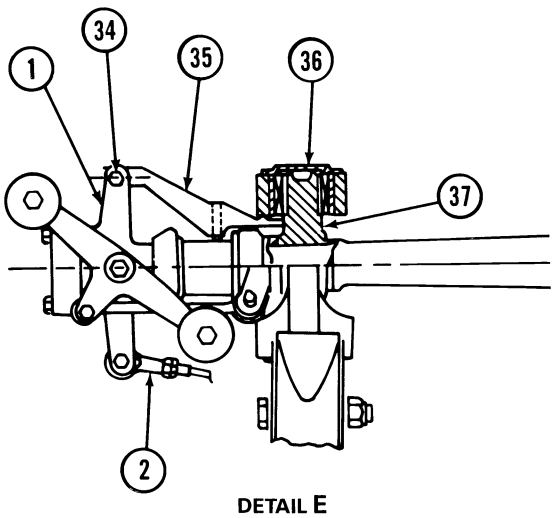
Figure 67-3. Antitorque controls rigging (Sheet 2)



DETAIL D
VIEW LOOKING DOWN



DETAIL F



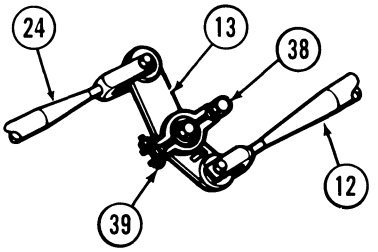
DETAIL E

NOTES

1. Remove one pitch link (2).
2. Position rigging tool (35) on crosshead (1) and trunnion (37) and install bolt (34).
3. Push inboard on crosshead (1) to cause tool to make firm contact with trunnion (37).
4. After rigging is completed, remove rigging tool (35) from tail rotor (36).
5. Install pitch link (2) that was removed in note 1.

412-SAR-67-3-3

Figure 67-3. Antitorque controls rigging (Sheet 3)



DETAIL G

1. Crosshead
2. Pitch links
3. Lever
4. Link
5. Control tube
6. Bellcrank
7. Hydraulic cylinder
8. Support
9. Bellcrank
10. Servoactuator
11. Lever
12. Control tube
13. Walking beam
14. Force gradient
15. Rotary actuator
16. Control tube
17. Copilot pedal adjuster
18. Copilot left pedal
19. Copilot right pedal
20. Transducers
21. Pilot pedal adjuster
22. Pilot left pedal
23. Pilot right pedal
24. Control tube
25. Bellcrank
26. Control tube
27. Tool assembly (412-201-701-101)
28. Bolt (20-057-4-18D)
29. Tool assembly (412-201-700-101)
30. Indexing pin
31. Bolt (20-057-4-23D)
32. Antirotation screws
33. Output arm
34. Bolt
35. Tail rotor rigging tool (T103206)
36. Tail rotor
37. Tail rotor hub trunnion
38. Clamp
39. Screw

412-SAR-67-3-4

Figure 67-3. Antitorque controls rigging (Sheet 4)

COLLECTIVE FLIGHT CONTROLS — DAFCS

67-8. COLLECTIVE CONTROLS — DAFCS.

67-9. DESCRIPTION.

The collective controls installation (figure 67-4) consists of the control linkage and components from pilot and copilot collective sticks (9 and 13) up to but not including collective lever (1).

The collective controls installation consists of a force gradient (18, figure 67-4), lever (17), bellcrank (19), link (20), support (21), electromechanical rotary actuator (22), and two transducers (14). The two transducers (14) are connected to the arm of jackshaft tube (11) and support.

The collective trim system consists of an electromechanical rotary actuator (22) and a force gradient (18) with a detent switch connected in parallel to the primary collective controls. The rotary actuator (22) repositions the collective controls in response to a signal from either the pilot or the AFCS system. Actuation of the force gradient (18) detent switch either by the rotary actuator driving to its internal stop or by a pilot control input will disengage the rotary actuator.

Two dual element linear transducers (14) connected in parallel to the collective controls are used to convert mechanical movement to an electrical position signals for the DAFCS and flight data recorder.

The cylinder (5) reduces effort required for control and reduces feedback forces from the main rotor.

Actuation of the pilot collective stick (13) or copilot collective stick (9) causes hydraulic power assisted collective pitch change of the main rotor blades.

Collective dual controls are installed for copilot control of the helicopter.

Operation of collective dual controls system is basically the same as operation of single system. Refer to Flight Manual for operating procedures.

For removal and installation instructions of copilot collective control system refer to BHT-412-SI-13 Refer to BHT-412-IPC for illustrated parts list.

67-10. COLLECTIVE LEVER.

67-11. Removal.

1. Remove access panel from cabin floor forward of pylon.

2. Remove bolt (3, figure 67-5).

3. Remove bolt (8).

4. Remove bolt (19).

5. Remove bolt (25).

6. Remove lever (17) from airframe structure.

67-12. Inspection.

1. Inspect lever (17, figure 67-5) for mechanical and corrosion damage in accordance with BHT-412-CR&O-1 manual.

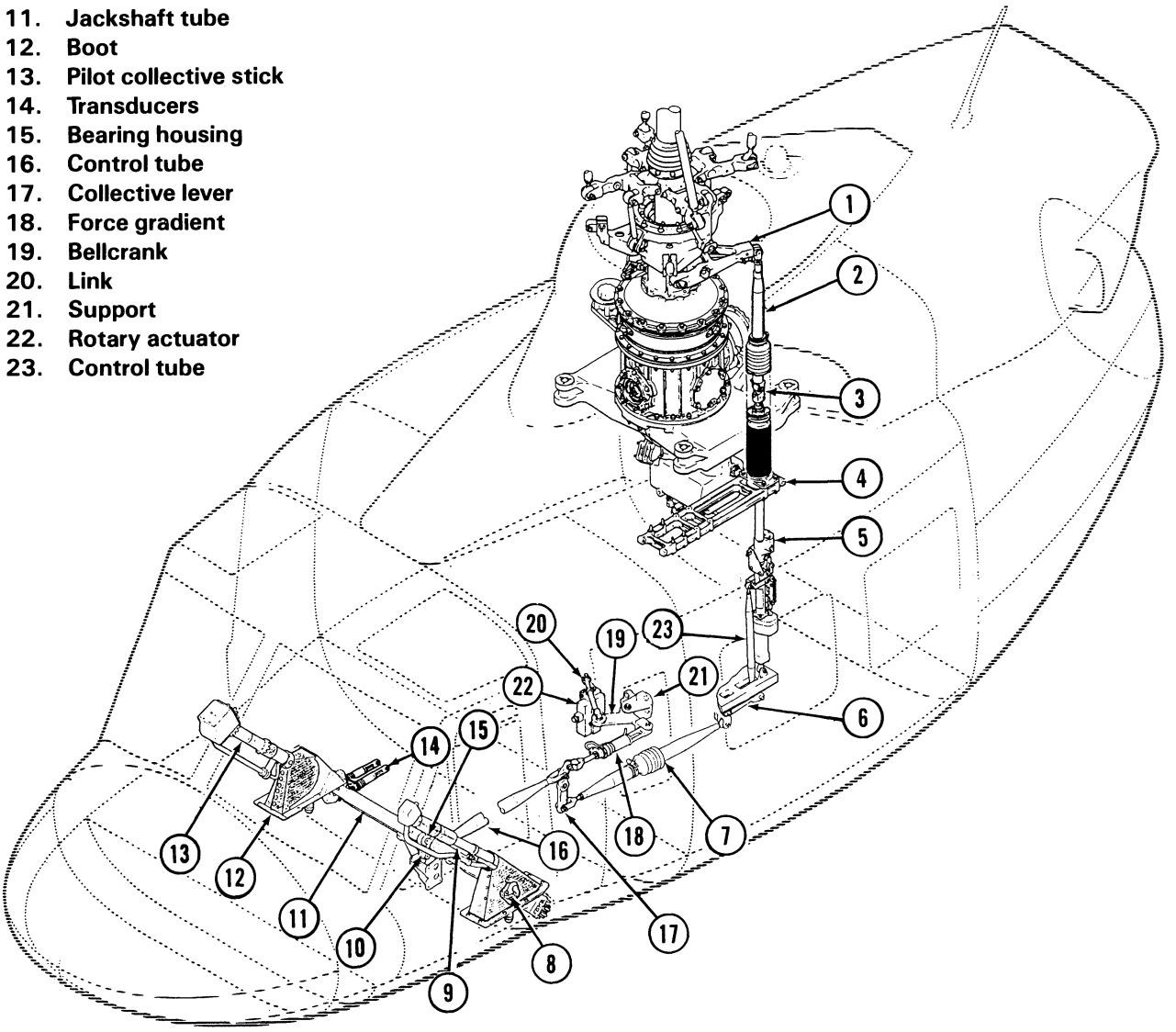
2. Inspect bearings in lever (17) for secure installation in lever and for wear (BHT-ALL-SPM).

67-13. Installation.

1. Place lever (17, figure 67-5) into position in airframe structure with long end down. Install bolt (19), aluminum washers (18 and 16), nut (15), and cotter pin (14).

2. Position adjustable control tube (10) on lever (17) and install bolt (8), steel washers (9 and 11), nut (12), and cotter pin (13).

1. Collective lever
2. Control tube
3. Universal
4. Support
5. Cylinder
6. Bellcrank
7. Adjustable control tube
8. Clamp
9. Copilot collective stick
10. Control arm
11. Jackshaft tube
12. Boot
13. Pilot collective stick
14. Transducers
15. Bearing housing
16. Control tube
17. Collective lever
18. Force gradient
19. Bellcrank
20. Link
21. Support
22. Rotary actuator
23. Control tube



412-SAR-67-4

Figure 67-4. Collective controls

3. Position control tube (2) on lever (17) and install bolt (3), steel washers (4 and 5), nut (6) and cotter pin (7).

4. Position force gradient (20) on lever (17) and install bolt (25), aluminum washers (24 and 23), nut (22), and cotter pin (21).

5. Move pilot collective stick (1) through full throw and check for binding and/or obstruction of controls.

6. Confirm that collective controls rigging is within limits. Refer to paragraph 67-5 .

7. Install access panel in cabin floor forward of pylon.

67-14. COLLECTIVE FORCE GRADIENT.

67-15. Removal.

1. Remove access panel from cabin floor forward of pylon.

2. Disconnect electrical connector (1, figure 67-6).

3. Remove bolt (2).

4. Remove bolt (13).

5. Remove force gradient (8).

67-16. Inspection.

1. Inspect force gradient (8, figure 67-6) for corrosion and mechanical damage BHT-412-CR&O-1 . Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is not acceptable.

2. Inspect bearing for secure installation in housing and for wear (BHT-ALL-SPM).

67-17. Repair.

1. Repair of force gradient consists of replacement of unservicable components and adjustment of spring preload in accordance with paragraph 67-18., step 1.

67-18. Installation.

MATERIALS REQUIRED

NUMBER	NOMENCLATURE
C-405	Lockwire

CAUTION

ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION ON ROTARY ACTUATOR.

NOTE

The collective force gradient is similar in appearance to the force gradients used in the cyclic controls but has a different part number. Ensure that correct part number force gradient is installed for the collective controls system.

1. If not previously accomplished, adjust spring preload on force gradient (8, figure 67-6, sheet 2), 40 to 41 lbs. (177.92 to 182.368 N) as follows:

a. Loosen jamnut (23) and remove clevis (15).

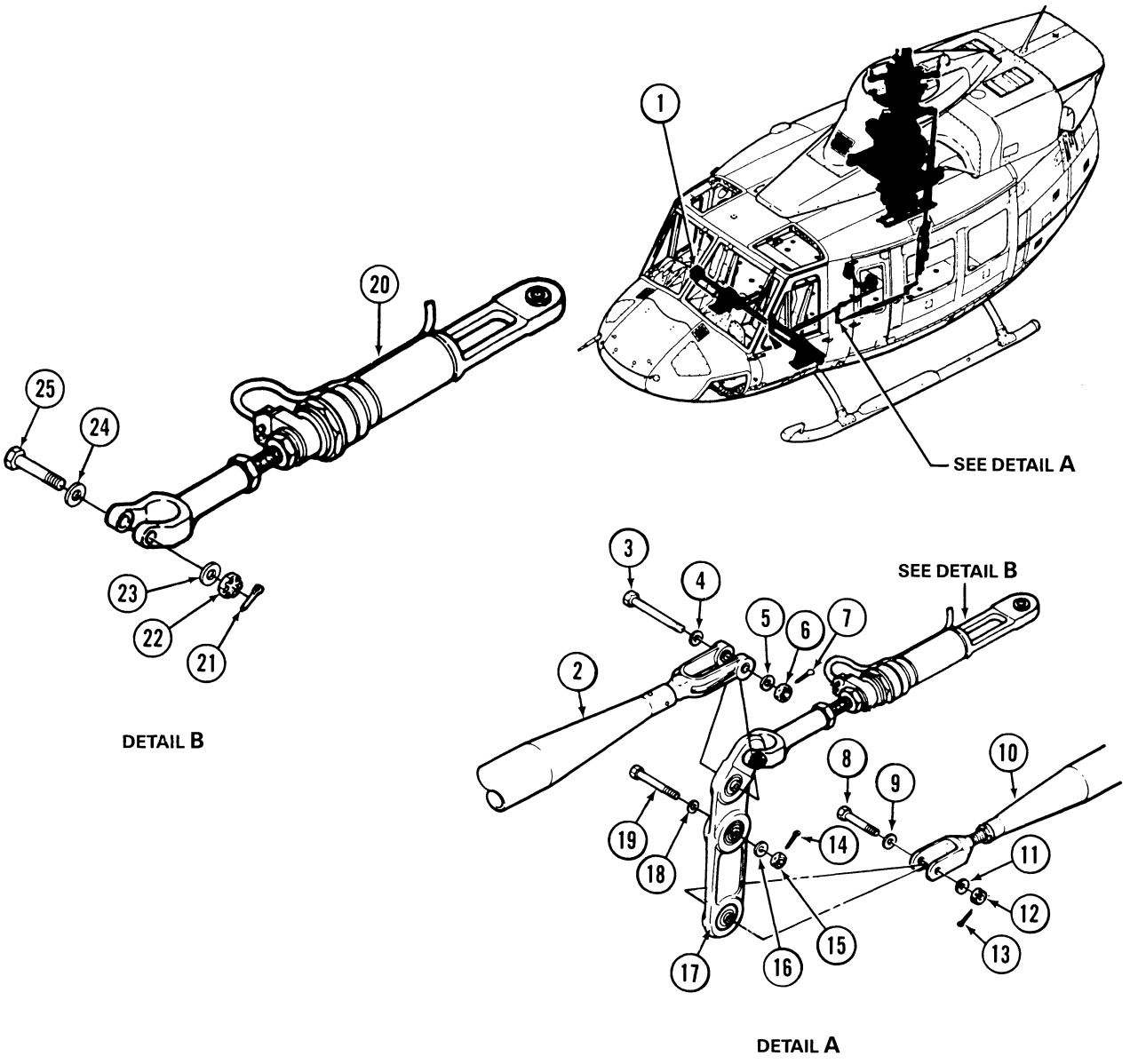
b. Cut straps (33) holding switch and cable assembly (30) to force gradient (8).

c. Remove nut (28), washer (27), screw (26), and packing (29), from cap (24), and switch and cable assembly (30).

d. Cut lockwire and loosen jamnut (31), remove cap (24) from cylinder (17).

e. Remove spring assembly (32) from cylinder (17).

f. Loosen jamnut (21) while holding shaft (22) at wrench flat. Adjust spring preload to 40 to 41 lbs. (177.92 to 182.368 N) by turning spacer (20), (using wrench flat) clockwise



1. Pilot collective stick
2. Control tube
3. Bolt
4. Aluminum washer
5. Aluminum washer
6. Nut
7. Cotter pin
8. Bolt
9. Steel washer
10. Adjustable control tube
11. Steel washer
12. Nut
13. Cotter pin
14. Cotter pin
15. Nut
16. Aluminum washer
17. Lever
18. Aluminum washer
19. Bolt
20. Force gradient
21. Cotter pin
22. Nut
23. Steel washer
24. Steel washer
25. Bolt

412-SAR-67-5

Figure 67-5. Collective lever removal/installation

against spring assembly (32). Tighten jamnut (21) against spacer (20).

g. Position spring assembly (32) in cylinder (17) and install cap (24) until all end play is eliminated. Torque jamnut (31) to 100 in. lbs. (11.298 Nm) and lockwire (C-405) to prevent turning in either direction. After assembly verify spring preload of 40 to 41 lbs. (177.92 to 182.368 N).

h. Install packing (29), and switch and cable assembly (30), on cap (24).

i. Install screws (26), washers (27), nuts (28), in cap (24) and switch and cable assembly (30). Do not tighten screws at this time.

j. Install clevis (15) on shaft (22). Adjust length of force gradient (8) as shown in figure 67-6, sheet 2. Tighten jamnut (23) to clevis (15).

k. Push and hold switch and cable assembly (30), and ball (25) against cap (24). Hold this position to iniatially adjust switch and cable assembly (30), then adjust switch to actuate within ± 0.050 in. (1.27 mm) linear shaft travel of spring assembly (32). Secure screws (26) mounting switch and cable assembly (30) to cap (24).

l. Install straps (33) holding switch and cable assembly (30) to force gradient (8).

2. Place force gradient (8) into position on bellcrank (3). Install bolt (2), steel washers (3 and 5), nut (6), and cotter pin (7).

3. Place force gradient (8) into position on lever (12). Install bolt (13), steel washers (14 and 11), nut (10), and cotter pin (9).

4. Connect electrical connector (1) at fuselage bracket.

5. Install access panel in cabin floor forward of pylon.

67-19. COLLECTIVE BELLCRANK.

67-20. Removal.

1. Remove access panel from cabin floor forward of pylon.

2. Remove bolt (1, figure 67-7).

3. Remove bolt (10).

4. Remove bolt (5).

5. Remove bellcrank (15) from support (4).

67-21. Inspection.

1. Inspect bellcrank (15, figure 67-7) for mechanical and corrosion damage in accordance with BHT-412-CR&O-1 manual.

2. Inspect bearings in bellcrank (15) for secure installation in bellcrank and for wear (BHT-ALL-SPM).

67-22. Installation.

1. Place bellcrank (15, figure 67-7) into position on support (4). Install bolt (5), aluminum washers (6 and 7), nut (8), and cotter pin (9).

2. Position force gradient (16) on bellcrank (15) and install bolt (10), aluminum washers (11 and 14), nut (13), and cotter pin (12).

3. Position link (3) on bellcrank (15) and install bolt (1), steel washers (2 and 19), nut (18) and cotter pin (17).

4. Install access panel in cabin floor forward of pylon.

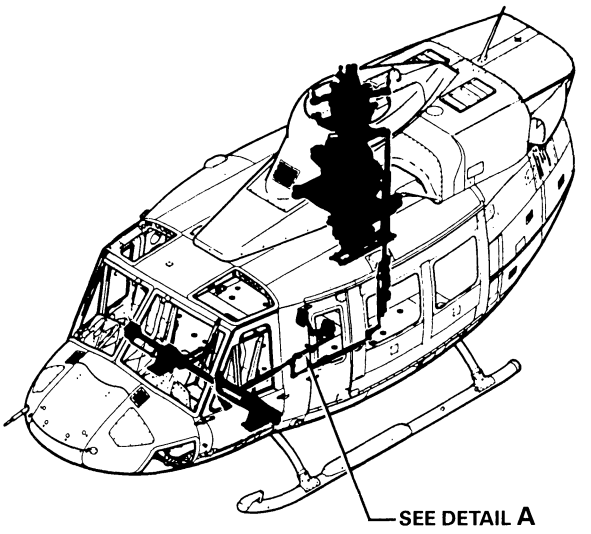
67-23. COLLECTIVE LINK.

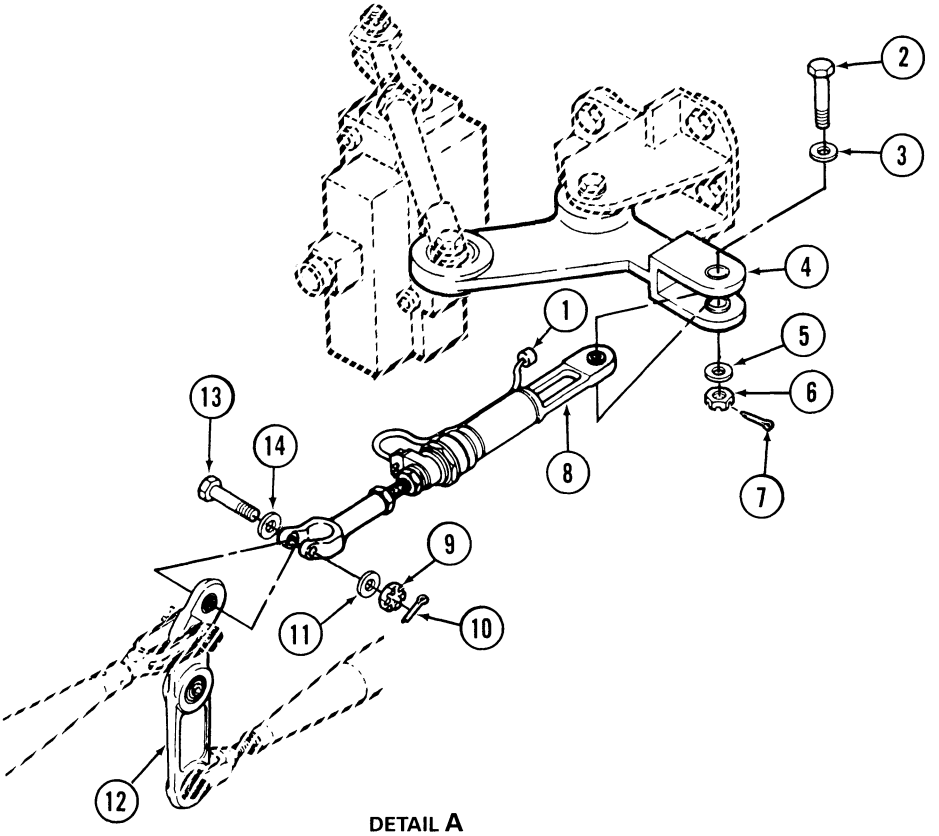
67-24. Removal.

1. Remove access panel from cabin floor forward of pylon.

2. Remove bolt (1, figure 67-8).

1. Electrical connector
2. Bolt
3. Aluminum washer
4. Bellcrank
5. Aluminum washer
6. Nut
7. Cotter pin
8. Force gradient
9. Cotter pin
10. Nut
11. Washer
12. Lever
13. Bolt
14. Washer

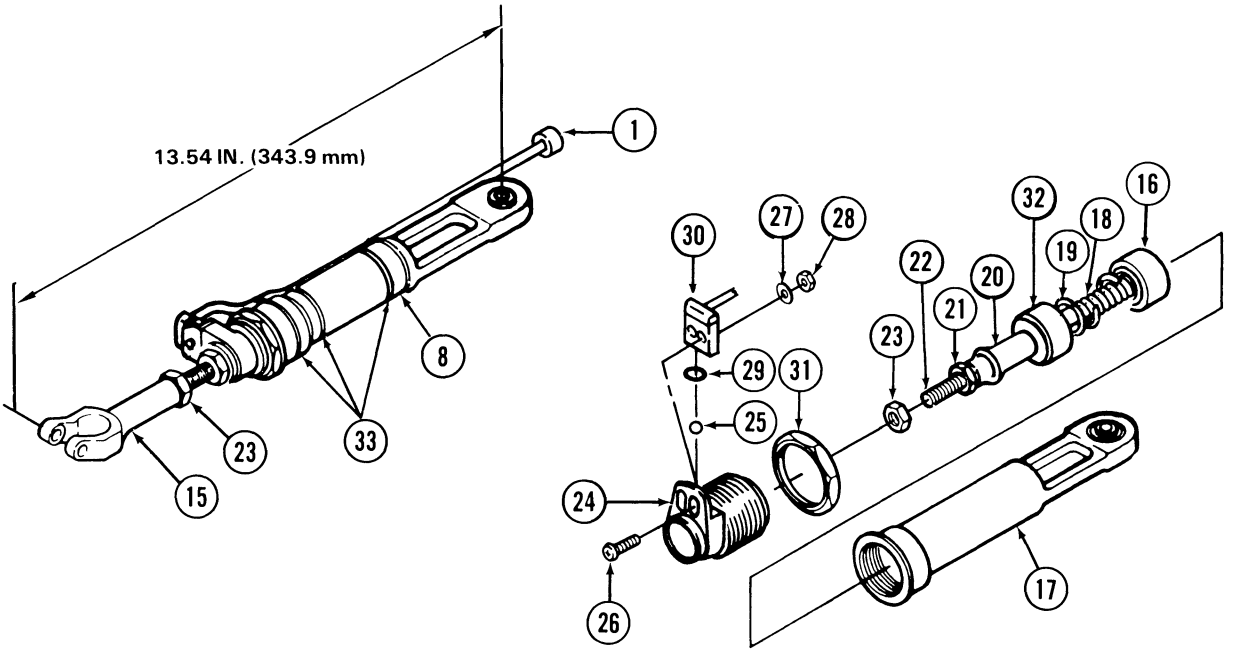




DETAIL A

412-SAR-67-6-1

Figure 67-6. Collective force gradient removal/installation (Sheet 1 of 2)



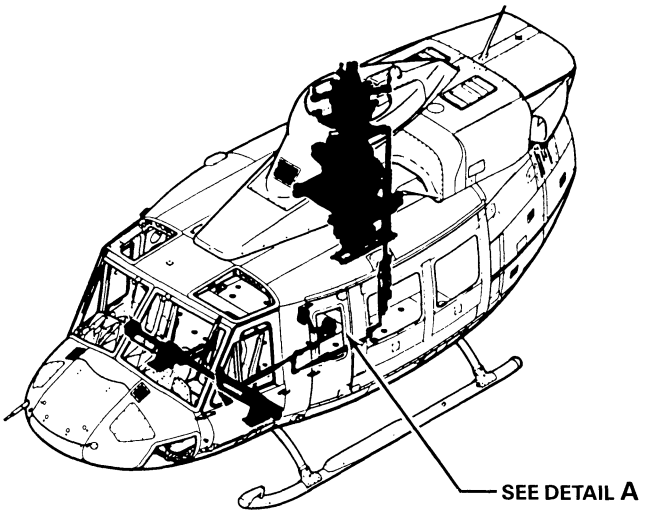
DETAIL B
VIEW LOOKING DOWN

15. Clevis
16. Guide
17. Cylinder
18. Spring
19. Spring
20. Spacer
21. Jamnut
22. Shaft
23. Jamnut
24. Cap
25. Ball
26. Screw
27. Washer
28. Nut
29. Packing
30. Switch and cable assembly
31. Jamnut
32. Spring assembly
33. Strap

412-SAR-67-6-2

Figure 67-6. Collective force gradient removal/installation (Sheet 2)

1. Bolt
2. Steel washer
3. Link
4. Support
5. Bolt
6. Aluminum washer
7. Aluminum washer
8. Nut
9. Cotter pin
10. Bolt
11. Aluminum washer
12. Cotter pin
13. Nut
14. Aluminum washer
15. Bellcrank
16. Force gradient
17. Cotter pin
18. Nut
19. Steel washer



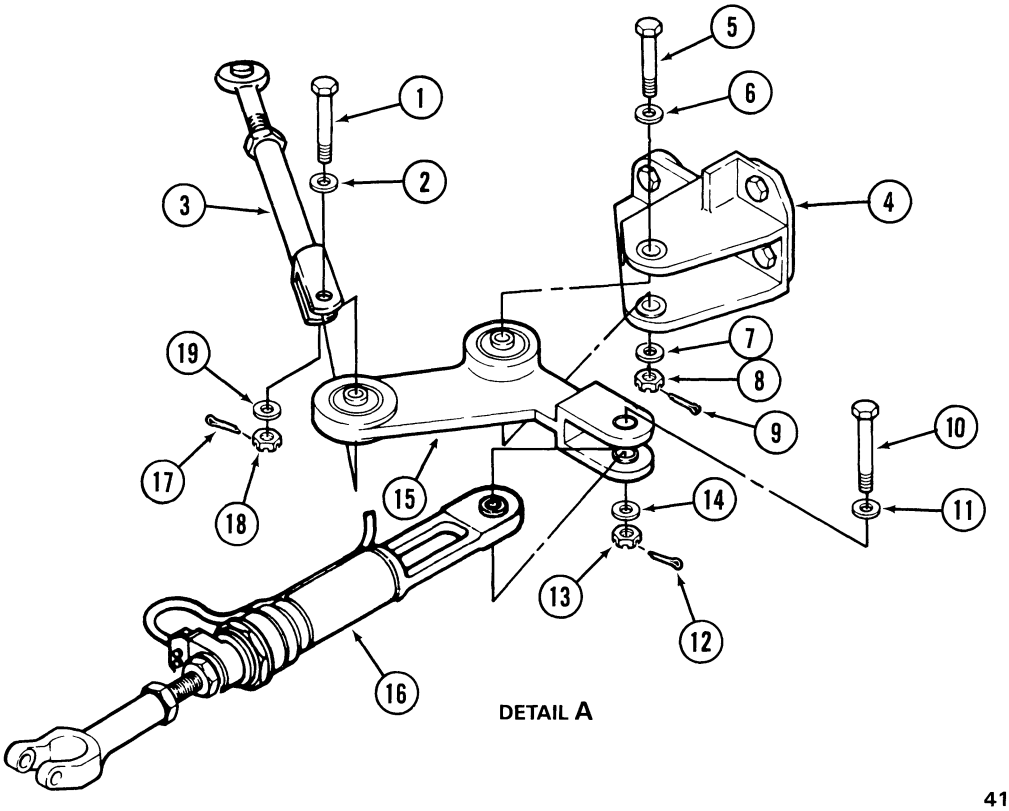


Figure 67-7. Collective bellcrank removal/installation

3. Remove nut (5).

4. Remove link (2) from bellcrank (7) and rotary actuator arm stud (3).

67-25. Inspection.

1. Inspect link (2, figure 67-8) for mechanical and corrosion damage in accordance with BHT-412-CR&O-1 manual.

2. Inspect bearing in link (2) for security and for wear BHT-ALL-SPM).

67-26. Installation.

1. Place link (2, figure 67-8) into position on bellcrank (7). Install bolt (1), steel washers (10 and 11), nut (9), and cotter pin (8).

2. Position link (2) on rotary actuator arm stud (3). Install steel washer (6), nut (5), and cotter pin (4).

3. Install access panel in cabin floor forward of pylon.

67-27. COLLECTIVE ROTARY ACTUATOR.

67-28. Removal.

1. Remove access panel from cabin floor forward of pylon.

2. Remove nut (2, figure 67-9).

3. Disconnect electrical connector (6) from rotary actuator (7).

4. Remove bolts (9), four places.

5. Remove rotary actuator (7) from structure (5).

67-29. Inspection.

1. Inspect rotary actuator (7, figure 67-9) for mechanical and corrosion damage and wear (BHT-ALL-SPM).

2. Inspect rotary actuator arm stud (4) for damaged threads. No damage that can be detected visually is acceptable.

3. Inspect electrical receptacle for bent pins and damaged threads. Superficial damage is acceptable without repair.

67-30. Repair.

Refer to vendor manual for rotary actuator repair instructions.

67-31. Installation.

1. Place rotary actuator (7, figure 67-9) into position on structure (5). Install four aluminum washers (8), and four bolts (9).

2. Position link (10) on rotary actuator arm stud (4) and install steel washer (3), nut (2), and cotter pin (1).

3. Connect electrical connector (6) to rotary actuator (7).

4. Install access panel in cabin floor forward of pylon.

67-32. COLLECTIVE TRANSDUCERS.

67-33. Removal.

1. Remove access panel from cabin floor forward of pylon.

2. Remove bolt (6, figure 67-10).

3. Remove bolt (8).

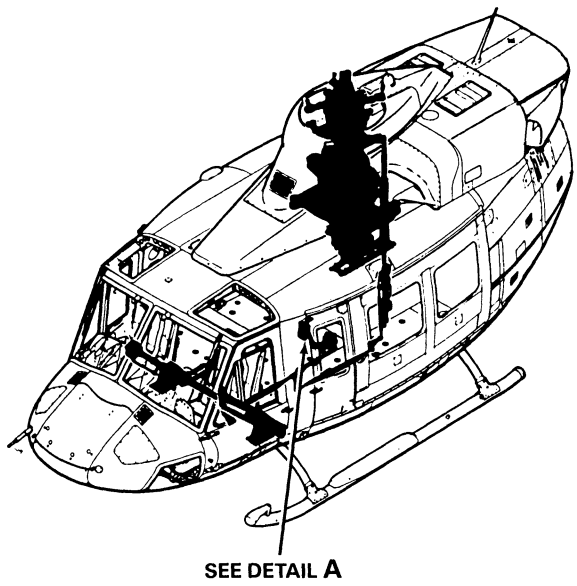
4. Disconnect electrical connectors from both transducers (7 and 15).

5. Remove transducers (7 and 15) from bracket (11) of collective jackshaft tube and support (4).

67-34. Inspection.

1. Inspect transducers (7 and 15, figure 67-10) for mechanical and corrosion damage.

1. Bolt
2. Link
3. Rotary actuator arm stud
4. Cotter pin
5. Nut
6. Steel washer
7. Bellcrank
8. Cotter pin
9. Nut
10. Steel washer
11. Steel washer



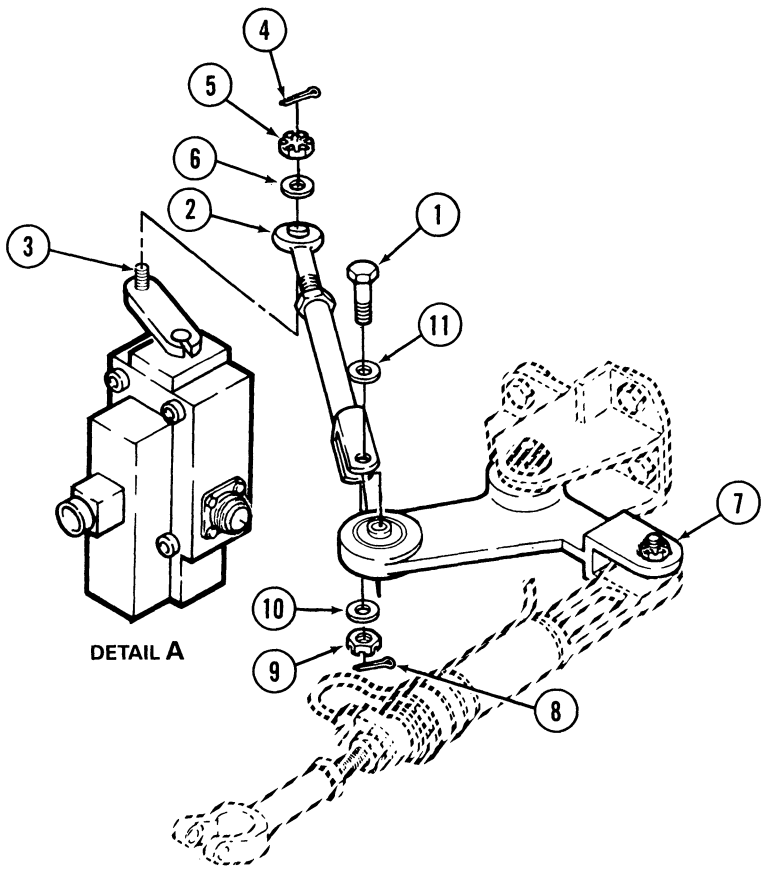
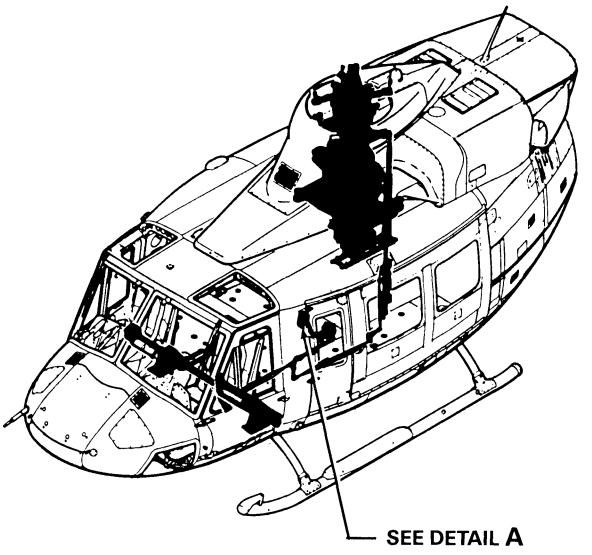
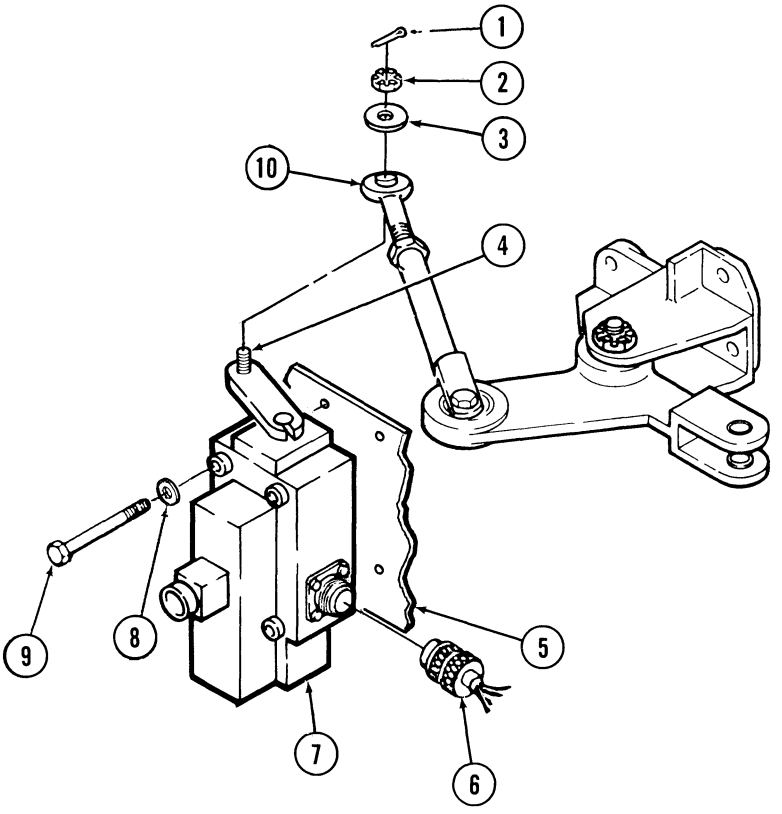


Figure 67-8. Collective link removal/installation

1. Cotter pin
2. Nut
3. Steel washer
4. Rotary actuator arm stud
5. Structure
6. Electrical connector
7. Rotary actuator
8. Aluminum washer
9. Bolt
10. Link





DETAIL A

412-SAR-67-9

Figure 67-9. Collective rotary actuator removal/installation

2. Inspect bearings in transducers (7 and 15) for secure installation in transducer and for wear (BHT-ALL-SPM).

3. Inspect electrical connector for bent pins and damaged threads. Superficial damage is acceptable without repair.

67-35. Repair.

1. Collective control transducers are nonreparable and must be replaced when defective.

67-36. Installation.

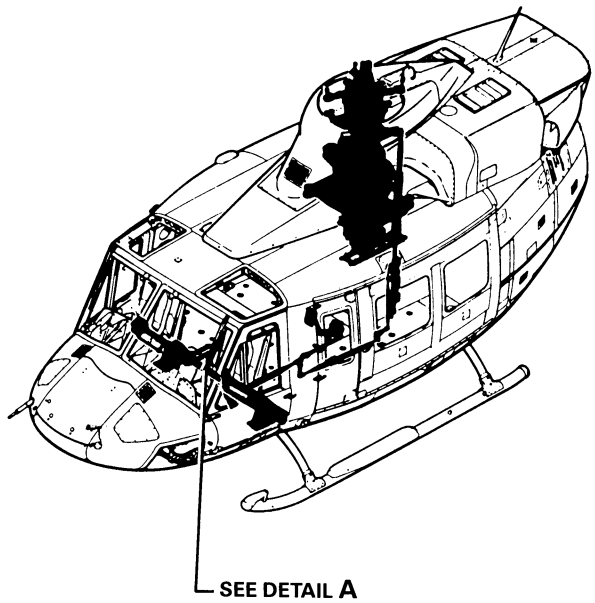
1. Position transducers (7 and 15, figure 67-10) on support (4). Install bolt (6), spacer (5), spacer (3) under nut (2), and cotter pin (1).

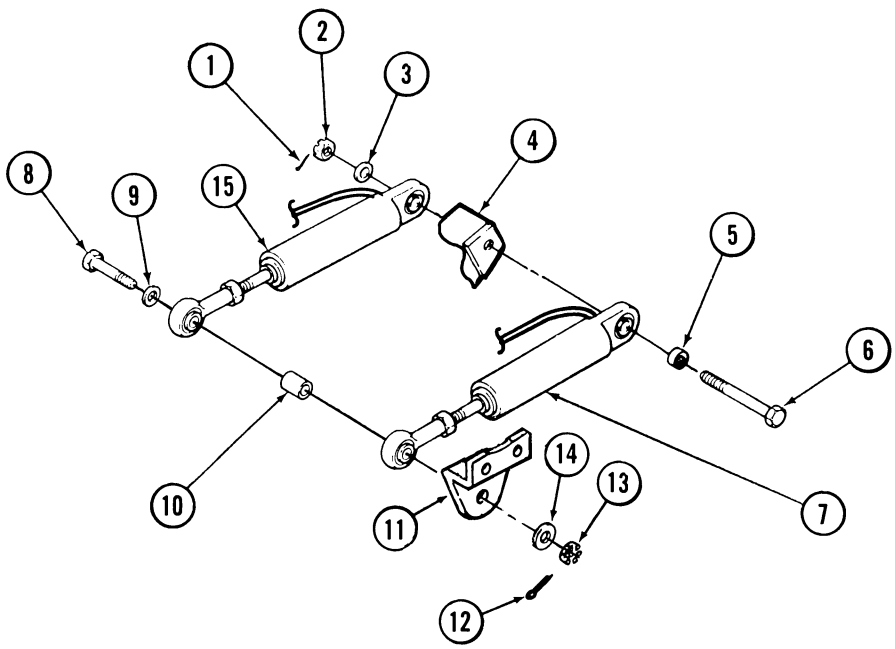
2. Place transducers (7 and 15) into position on bracket (11) of collective jackshaft lever. Install spacer (9) under head of bolt (8), spacer (10), spacer (14), nut (13), and cotter pin (12).

3. Connect electrical connectors to both transducers (7 and 15).

4. Install access panel in cabin floor forward of pylon.

1. Cotter pin
2. Nut
3. Spacer
4. Support
5. Spacer
6. Bolt
7. Transducer
8. Bolt
9. Spacer
10. Spacer
11. Bracket
12. Cotter pin
13. Nut
14. Spacer
15. Transducer





DETAIL A

412-SAR-67-10

Figure 67-10. Collective transducer removal/installation

CYCLIC CONTROLS — DAFCS

67-37. CYCLIC CONTROLS — DAFCS.

67-38. DESCRIPTION.

The cyclic controls installation (figure 67-11) consist of the controls and components from the pilot and copilot cyclic stick (33 and 15) up to but not including the swashplate (1).

The cyclic control system incorporates two servoactuators (14 and 35), one connected in roll system and one connected in pitch system. A control tube and idler (37) are attached to servoactuator (14) in the pitch system.

Four transducers, two lateral cyclic transducers (29) in roll system and two fore and aft cyclic transducers (34) in pitch system, are used to convert mechanical movement to electrical signals for injecting pilot and copilot control commands to the DAFCS.

Two fore and aft cyclic transducers (34) are attached to right end of cyclic jackshaft (16) providing a fore and aft cyclic signal. Two lateral cyclic transducers (29) are attached to a forward cyclic bellcrank (31) providing a lateral cyclic signal.

A force gradient assembly connected to an electromechanical rotary actuator is also installed in each cyclic system. Cyclic roll system force gradient attaches to a bellcrank forward of jackshaft. The cyclic pitch system force gradient attaches directly to the left end of cyclic jackshaft.

A support assembly and idler (37) are used to connect the fore and aft cyclic servoactuator (14) and a pitch control tube which is attached to the mixing lever assembly (39).

Two proximity switches (36 and 38), and two cyclic centering (CYC CTR) caution lights are installed for aid in centering cyclic sticks. The lights are located in the pilots and copilots instrument panel. The two proximity switches (36 and 38) are located underneath the floor near the mixing lever assembly (39).

NOTE

Cyclic control DAFCS system components forward of mixing lever assembly (39) are identified as ‘lateral cyclic‘ or ‘fore and aft cyclic‘ as applicable, i.e., ‘lateral cyclic force gradient‘.

Cyclic control system components aft of mixing lever assembly (39) are identified as ‘left cyclic‘ or ‘right cyclic‘ as applicable, i.e., ‘left cyclic flight control cylinder.

The fore and aft control linkage and the lateral control linkage are separate from the cyclic stick to the mixing lever assembly (39). From the mixing lever assembly (39) to the swashplate (1) the effects of the control linkages are combined and cannot be considered separately as to effect.

Actuation of the pilot and copilot cyclic sticks (33 and 15) causes hydraulic power assisted cyclic pitch change of the main rotor blades.

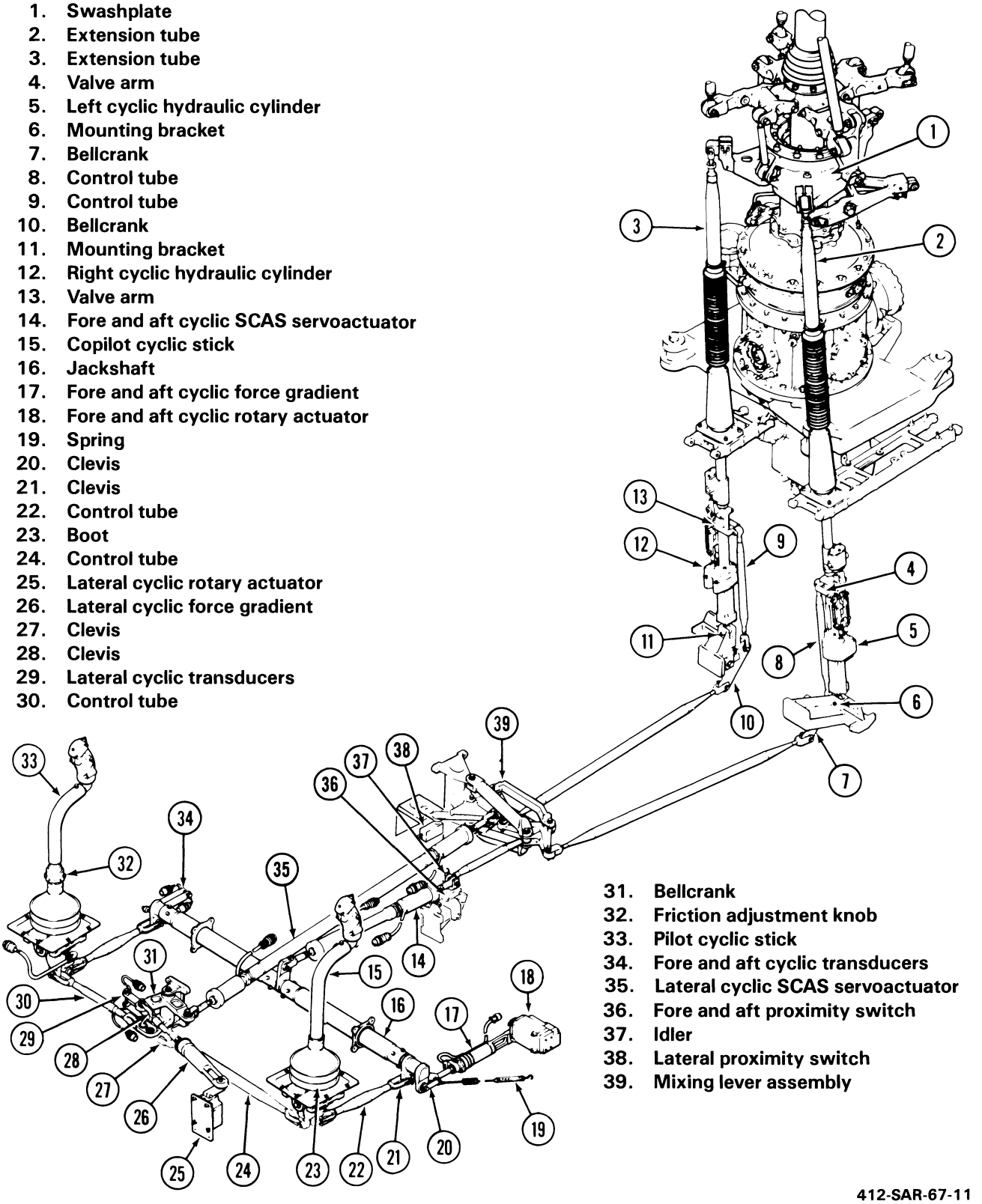
The forward actuators in each cyclic system are connected to Helipilot (HP-1) system. Aft actuators in each cyclic system are connected to Helipilot (HP-2) system.

Refer to Chapter 98 , this supplement for DAFCS system wiring diagrams.

Dual controls are installed for copilot control of the helicopter.

Operation of dual controls system is basically the same as operation of single system. Refer to Flight Manual for operating procedures.

1. Swashplate
2. Extension tube
3. Extension tube
4. Valve arm
5. Left cyclic hydraulic cylinder
6. Mounting bracket
7. Bellcrank
8. Control tube
9. Control tube
10. Bellcrank
11. Mounting bracket
12. Right cyclic hydraulic cylinder
13. Valve arm
14. Fore and aft cyclic SCAS servoactuator
15. Copilot cyclic stick
16. Jackshaft
17. Fore and aft cyclic force gradient
18. Fore and aft cyclic rotary actuator
19. Spring
20. Clevis
21. Clevis
22. Control tube
23. Boot
24. Control tube
25. Lateral cyclic rotary actuator
26. Lateral cyclic force gradient
27. Clevis
28. Clevis
29. Lateral cyclic transducers
30. Control tube



31. Bellcrank
32. Friction adjustment knob
33. Pilot cyclic stick
34. Fore and aft cyclic transducers
35. Lateral cyclic SCAS servoactuator
36. Fore and aft proximity switch
37. Idler
38. Lateral proximity switch
39. Mixing lever assembly

412-SAR-67-11

Figure 67-11. Cyclic controls

67-39. ROTARY ACTUATOR (FORE AND AFT CYCLIC).

67-40. Removal.

1. Remove access panel from cabin floor under left crew seat. Remove access panel from bottom left side of helicopter just forward of forward landing gear crosstube.

2. Disconnect hook (14, figure 67-12) from bolt (10).

3. Remove nut (4) and separate force gradient (2) from arm (6).

4. Disconnect electrical connector (12).

5. Remove three bolts (9).

6. Remove one bolt (10).

7. Remove rotary actuator (7).

67-41. Inspection.

1. Inspect rotary actuator (7, figure 67-12) for mechanical and corrosion damage and wear (BHT-ALL-SPM).

2. Inspect rotary actuator arm (6) stud for damaged threads. No damage that can be detected visually is acceptable.

3. Inspect electrical receptacle for bent pins and damaged threads. Superficial damage is acceptable without repair.

67-42. Installation.

1. Inspect rotary actuator (7, 67-12) to ensure that arm (6) is installed on rotary actuator with index mark on arm positioned 14 teeth from index mark on rotary actuator shaft as shown on view A. Reposition arm (6) on rotary actuator if index marks are not properly located.

2. Position rotary actuator (7) on structure (13). Install special bolt (10) and aluminum washer (11) in lower outboard hole. Install three bolts (9) and aluminum washers (8) in remaining holes.

3. Position force gradient (2) on arm (6) stud. Install nut (4) and cotter pin (3).

4. Install spring (17), plate (15), and hook (14) on bolt (10). Refer to paragraph 67-51) for procedure to position spring (17) in correct hole in plate (15).

NOTE

For initial installation, install spring in fourth hole of plate (15). With hydraulic boost on, or during ground run, adjust spring tension to position pilot control stick (longitudinal cyclic only) perpendicular to deck within 2 degrees.

5. Close loop on hook (14) through bolt (10).

6. Install electrical connector (12).

7. Move cyclic controls through full travel and check for binding and /or obstructions.

8. Confirm that cyclic control rigging is within limits (BHT-412-MM , Chapter 67).

9. Install access panel on cabin floor under left crew seat.

10. Install access panel on bottom left side of helicopter just forward of forward landing gear crosstube.

67-43. ROTARY ACTUATOR (LATERAL CYCLIC).

67-44. Removal.

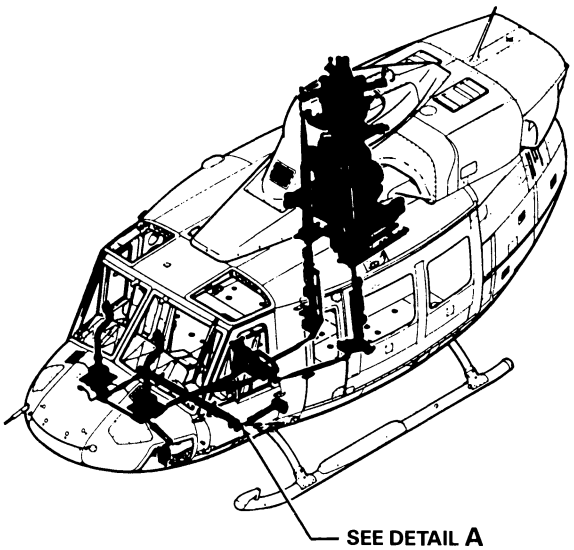
1. Remove access panel from bottom of helicopter just aft and inboard of left lower nose window.

2. Remove nut (3, 67-13).

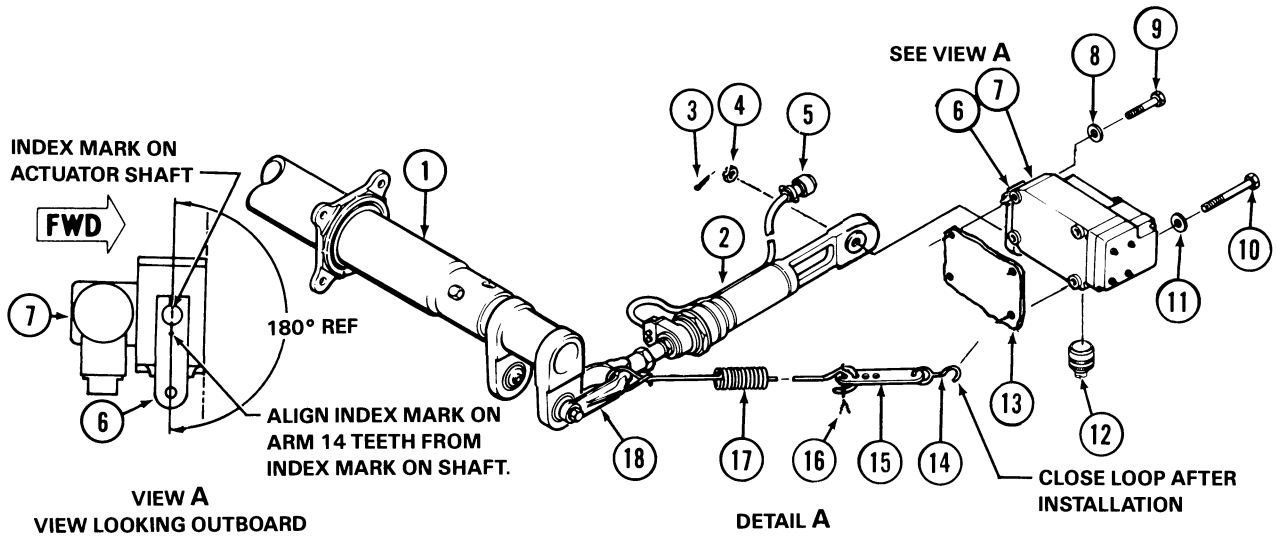
3. Disconnect electrical connector (9) from rotary actuator (7).

4. Remove four bolts (5).

5. Remove rotary actuator (7).



1. Tube and lever assembly
2. Force gradient
3. Cotter pin
4. Nut
5. Electrical connector
6. Arm
7. Rotary actuator
8. Aluminum washer
9. Bolt
10. Bolt (212-001-315-1)
11. Aluminum washer
12. Electrical connector
13. Structure
14. Hook
15. Plate
16. Lockwire
17. Spring
18. Clevis



412-SAR-67-12

Figure 67-12. Fore and aft cyclic rotary actuator removal/installation

67-45. Inspection.

1. Inspect rotary actuator (7, figure 67-13) for mechanical and corrosion damage and wear (BHT-ALL-SPM).

2. Inspect rotary actuator arm (10) stud for damaged threads. No damage that can be detected visually is acceptable.

3. Inspect electrical receptacle for bent pins and damaged threads. Superficial damage is acceptable without repair.

67-46. Installation.

1. Inspect rotary actuator (7, figure 67-13) to ensure that arm (10) is installed on rotary actuator with index mark on arm positioned six teeth from index mark on rotary actuator shaft as shown in view A. Reposition arm (10) on rotary actuator if index marks are not properly located.

2. Position rotary actuator (7) on rotary actuator adapter plate (structure) (8). Install four bolts (5) and four aluminum washers (6).

3. Position force gradient (2) on arm (10). Install nut (3) and cotter pin (4).

4. Install electrical connector (9).

5. Move cyclic controls through full throw and check for binding and/or obstructions.

6. Confirm that cyclic control rigging is within limits (BHT-412-MM , Chapter 67).

7. Install access panel on bottom of helicopter just aft and inboard of left lower nose window.

67-47. FORCE GRADIENT (FORE AND AFT CYCLIC).

67-48. Removal.

1. Remove access panel from cabin floor under left crew seat.

2. Remove access panel from bottom of helicopter on left side forward of forward landing gear crosstube.

3. Remove hook (7, figure 67-14) from bolt (6).

4. Disconnect electrical connector.

5. Remove nut (4).

6. Remove bolt (12).

7. Remove force gradient (2).

67-49. Inspection.

1. Inspect force gradient (2, figure 67-14) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion is not acceptable.

2. Inspect clevis (13) for damaged threaded helicoil insert and for elongation hole in bushing opposite the threaded helicoil insert.

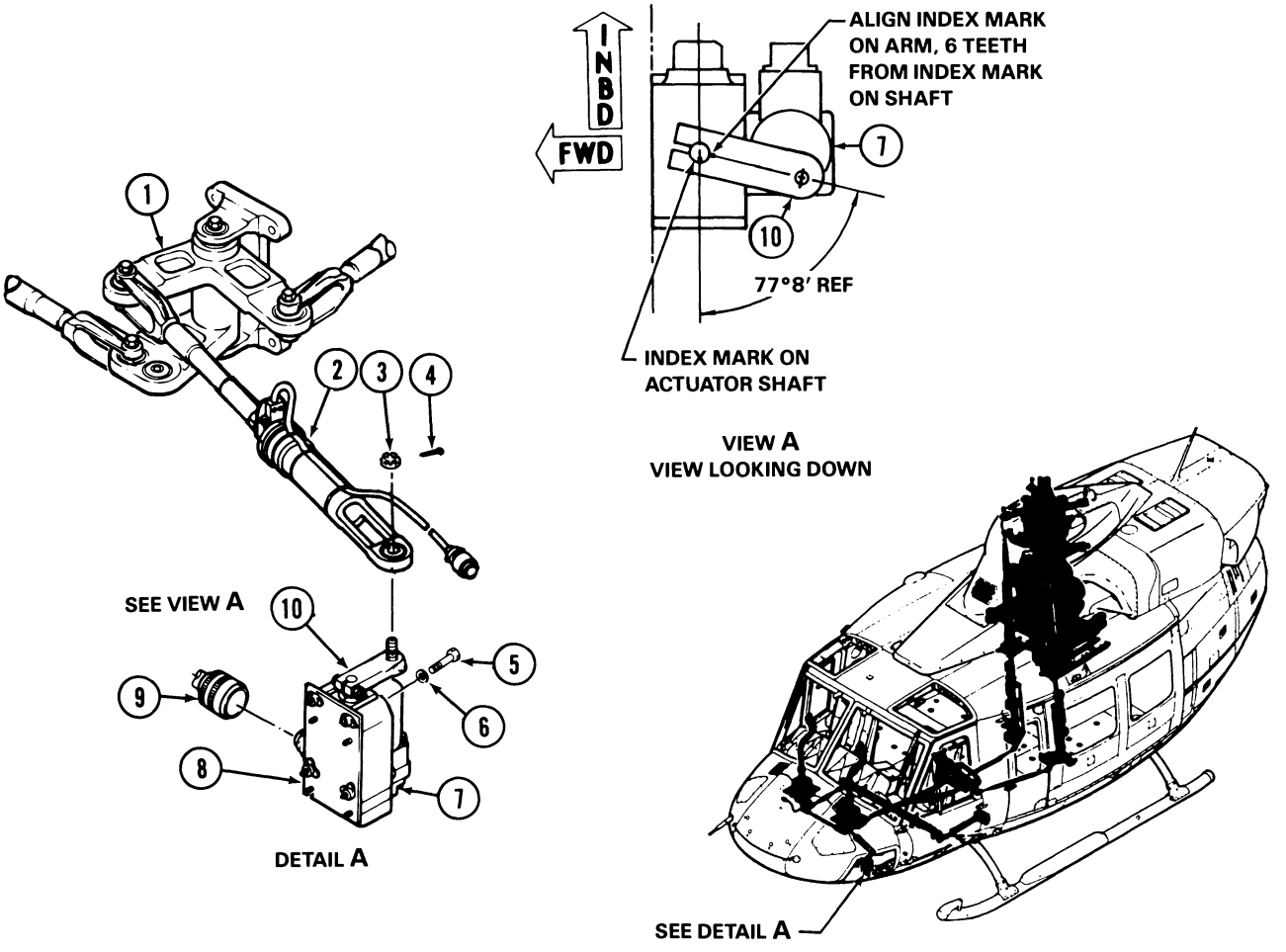
3. Inspect hook (7), plate (8), and spring (10) for obvious mechanical damage and for loss of tension in spring. Spring tension at 10.36 in. (263.1 mm) is 6.4 ± 0.5 lbs. (2.9 ± 0.2 kg).

4. Inspect bearing for wear and for secure installation in cylinder (5, figure 67-15). Refer to Bearing Wear Limit Table in BHT-ALL-SPM for acceptable limits.

5. Inspect two nuts (6 and 8) for secure installation. If adjustment of nuts (2 and 3) (spring preload) is suspected of being incorrect, adjust spring preload as outlined in paragraph 67-51 , step 1.

6. Inspect nut (1) and cylinder (5) for correct lockwire installation.

7. Inspect switch and electrical cable assembly (14) for frayed wires and broken pins in connector. Check continuity from connector pins through switch.



1. Lateral control bellcrank
2. Force gradient
3. Nut
4. Cotter pin
5. Bolt
6. Aluminum washer
7. Rotary actuator
8. Rotary actuator adapter plate (structure)
9. Electrical connector
10. Arm

412-SAR-67-13

Figure 67-13. Lateral cyclic rotary actuator removal/installation

67-50. Repair.

Repair of force gradient consists of replacement of unservicable components and adjustment of spring preload in accordance with paragraph 67-51., step 1.

67-51. Installation.

MATERIALS REQUIRED

NUMBER	NOMENCLATURE
C-405	Lockwire

NOTE

Force gradient (2, figure 67-14) in the fore and aft cyclic controls is similar in appearance to the force gradient in the lateral cyclic controls, but the force gradients have different part numbers. Ensure that correct part number force gradient is installed in the fore and aft position.

1. If not previously accomplished, adjust spring preload on force gradient as follows:

a. Remove cap (9, figure 67-15).

b. Remove spring assembly (4) from cylinder (5).

c. Adjust inboard nut (3) to provide 9 to 10 lbs. (40.032 to 44.48 N) preload against spring assembly (4). Tighten outboard nut (2) against inboard nut (3).

d. Position spring assembly (4) in cylinder (5) and install cap (9) until all end play is eliminated. Install jamnut (1) and lockwire (C-405) to prevent turning in either direction.

2. If not previously accomplished, adjust switch (14, figure 67-15) on force gradient as follows:

a. Loosen nut (8) on shaft of spring assembly (4).

b. Align actuator (7) on shaft of spring assembly (4) with ball (10) and plunger of switch (14). Secure actuator (7) with nut (8).

c. Loosen screws (13) mounting switch (14) to retainer (11). Push and hold actuator (7) sideways against cap (9) at point opposite location of switch (14). Hold this position to iniatially adjust switch (14) then adjust switch to actuate within ± 0.050 in. (1.27 mm) linear travel of spring assembly (4). Secure screws (13) mounting switch (14) to retainer (11).

3. Position spring (10, figure 67-14) on clevis (13).

4. Position force gradient (2) on tube and lever (1). Clevis (13) has a threaded helicoil insert in one side. Position the clevis with the threaded insert inboard. Install bolt (12) and aluminum washer (11). Lockwire (C-405) bolt head to hole in clevis (13).

CAUTION

ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION OF FORCE GRADIENT ON ROTARY ACTUATOR.

5. Position force gradient (2) on rotary actuator (5). Install nut (4) and cotter pin (3).

6. Install plate (8) on spring (10) with spring through fourth hole in plate (8). Do not lockwire spring to plate at this time.

7. Attach plate (8) with hook (7) to bolt (6). Do not close loop on hook (7) at this time.

8. Connect electrical connector.

9. Move fore and aft cyclic controls through full throw and check for binding and obstructions.

10. Apply external hydraulic power to helicopter (BHT-412-MM , Chapter 29).

11. Loosen cyclic friction and observe cyclic stick position. If cyclic stick is not

perpendicular (longitudinally) within two degrees, move spring (10) to different hole in plate (8) to cause cyclic stick to be vertical (longitudinally).

12. When tension on spring (10) is correctly adjusted, secure spring (10) to plate (8) with lockwire (C-405) . Close loop on hook (7) to prevent accidental disengagement of hook.

13. Move cyclic controls through full throw and check for binding and obstructions.

14. Confirm that cyclic control rigging is within limits (BHT-412-MM , Chapter 67).

15. Remove external power from helicopter.

16. Install access panel on bottom of helicopter on left side forward of forward landing gear crosstube.

17. Install access panel on cabin floor under left crew seat.

67-52. FORCE GRADIENT (LATERAL CYCLIC).

67-53. Removal.

1. Remove access panel from bottom of helicopter below force gradient (5, figure 67-16).

2. Disconnect electrical connector.

3. Remove nut (6).

4. Remove bolt (2).

5. Remove force gradient (5).

67-54. Inspection.

1. Inspect force gradient (5, figure 67-16) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion is not acceptable.

2. Inspect clevis on link (4) for elongated holes.

3. Inspect bearing for wear and for secure installation in cylinder (5, figure 67-17). Refer to Bearing Wear Limit Table in BHT-ALL-SPM for acceptable limits.

4. Inspect two nuts (6 and 8) for secure installation. If adjustment of nuts (2 and 3) (spring preload) is suspected of being incorrect, adjust spring preload as outlined in paragraph 67-56., step 1.

5. Inspect nut (1) and cylinder (5) for correct lockwire installation.

6. Inspect switch and electrical cable assembly (14) for frayed wires and broken pins in connector. Check continuity from connector pins through switch.

67-55. Repair.

Repair of force gradient consists of replacement of unservicable components and adjustment of spring preload in accordance with paragraph 67-56.

67-56. Installation.

MATERIALS REQUIRED

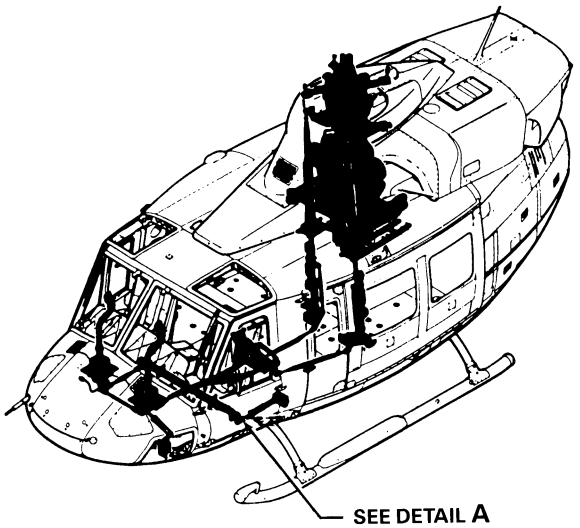
NUMBER	NOMENCLATURE
C-405	Lockwire

NOTE

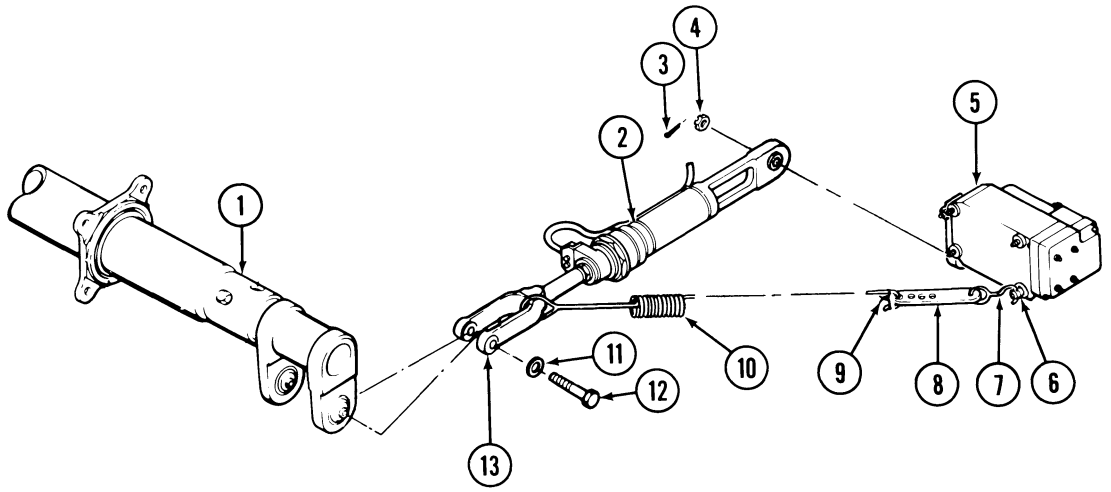
Force gradient (5, figure 67-16) in the lateral cyclic controls is similar in appearance to the force gradient in the fore and aft cyclic controls, but the force gradients have different part numbers. Ensure that correct part number force gradient is installed in the lateral position.

1. If not previously accomplished, adjust spring preload on force gradient as follows:

a. Remove cap (9, figure 67-17).



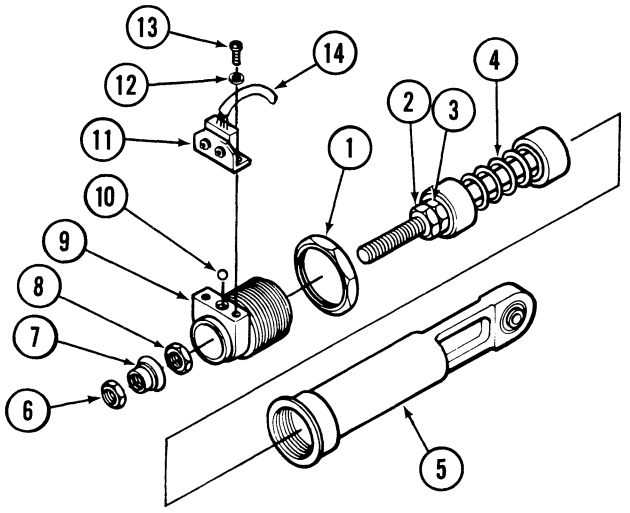
1. Tube and lever assembly
2. Fore and aft cyclic controls force gradient
3. Cotter pin
4. Nut
5. Rotary actuator
6. Bolt (212-001-315-1)
7. Hook
8. Plate
9. Lockwire
10. Spring
11. Aluminum washer
12. Bolt
13. Clevis



DETAIL A

412-SAR-67-14

Figure 67-14. Fore and aft cyclic force gradient removal/installation



1. Jamnut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Retainer
12. Washer
13. Screw
14. Switch and cable assembly

412-SAR-67-15

Figure 67-15. Fore and aft cyclic force gradient

b. Remove spring assembly (4) from cylinder (5).

c. Adjust inboard nut (3) to provide 6 to 7 lbs. (26.688 to 31.136 N) preload against spring assembly (4). Tighten outboard nut (2) against inboard nut (3).

d. Position spring assembly (4) in cylinder (5) and install cap (9) until all end play is eliminated. Install jamnut (1) and lockwire (C-405) to prevent turning in either direction.

2. If not previously accomplished, adjust switch (14, figure 67-17) on force gradient as follows:

a. Loosen nut (8) on shaft of spring assembly (4).

b. Align actuator (7) on shaft of spring assembly (4) with ball (10) and plunger of switch (14). Secure actuator (7) with nut (8).

c. Loosen screws (13) mounting switch (14) to retainer (11). Push and hold actuator (7) sideways against cap (9) at point opposite location of switch (14). Hold this position to iniatially adjust switch (14) then adjust switch to actuate within ± 0.050 in. (1.27 mm) linear travel of spring assembly (4). Secure screws (13) mounting switch (14) to retainer (11).

3. Position force gradient (5, figure 67-16) on lateral cyclic control bellcrank (1). Install bolt (2), aluminum washers (3 and 11), nut (10), and cotter pin (9).

CAUTION

ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION OF FORCE GRADIENT ON ROTARY ACTUATOR.

4. Position force gradient (5) on rotary actuator (8). Install nut (6) and cotter pin (7).

5. Connect electrical connector.

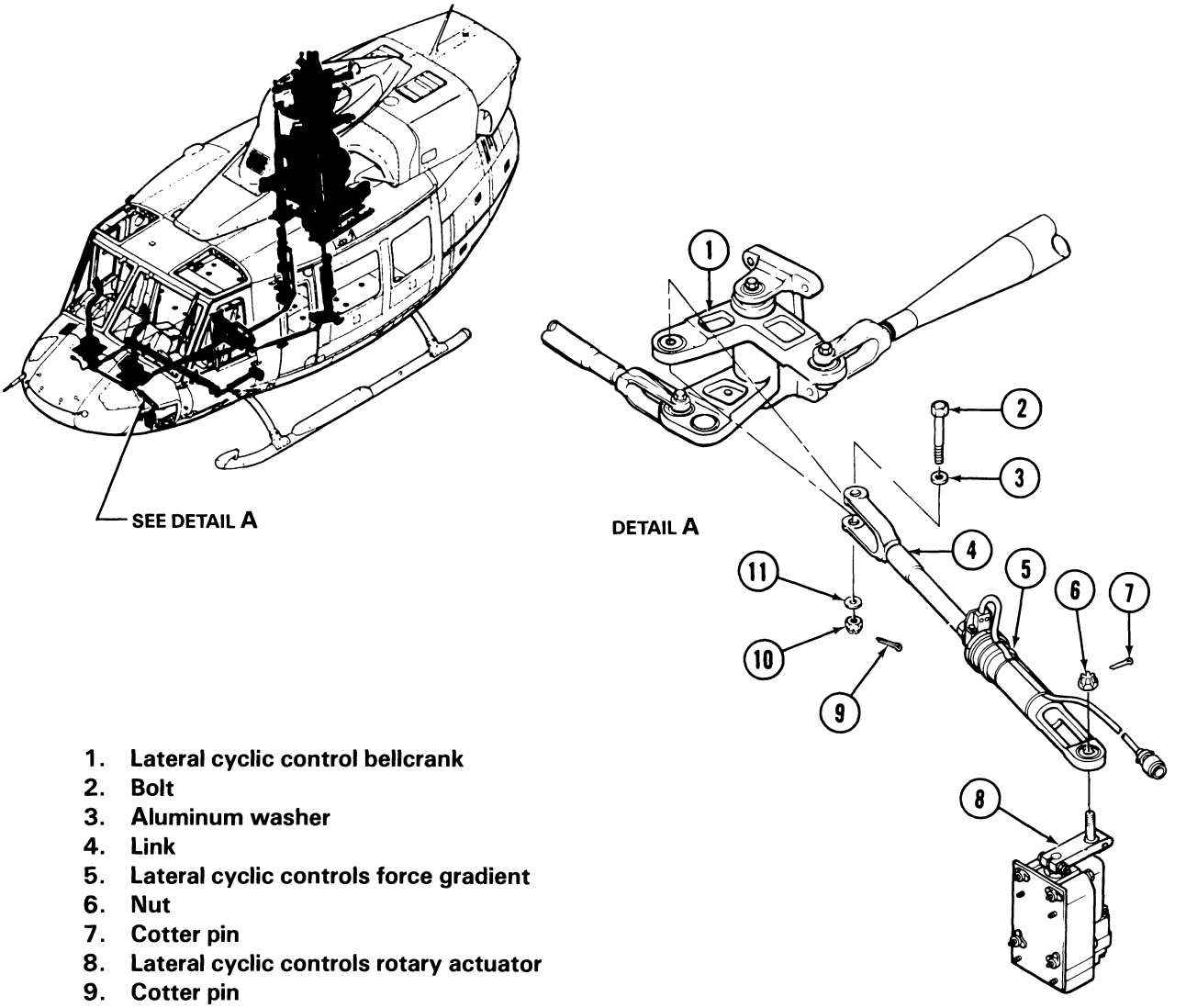
6. Apply external hydraulic power to helicopter (BHT-412-MM , Chapter 29).

7. Move cyclic controls through full throw and check for binding and/or obstructions.

8. Verify that cyclic control rigging is within limits (BHT-412-MM , Chapter 67).

9. Remove external power from helicopter.

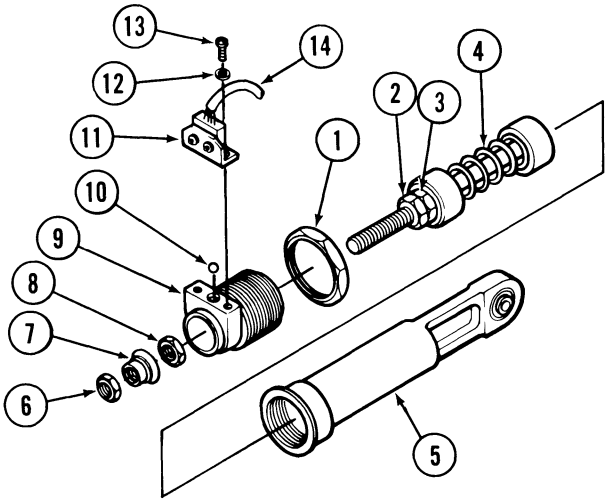
10. Install access panel on bottom of helicopter below force gradient.



1. Lateral cyclic control bellcrank
2. Bolt
3. Aluminum washer
4. Link
5. Lateral cyclic controls force gradient
6. Nut
7. Cotter pin
8. Lateral cyclic controls rotary actuator
9. Cotter pin
10. Nut
11. Aluminum washer

412-SAR-67-16

Figure 67-16. Lateral cyclic force gradient removal/installation



1. Jamnut
2. Outboard nut
3. Inboard nut
4. Spring assembly
5. Cylinder
6. Nut
7. Actuator
8. Nut
9. Cap
10. Ball
11. Retainer
12. Washer
13. Screw
14. Switch and cable assembly

412-SAR-67-17

Figure 67-17. Lateral cyclic force gradient

ANTITORQUE CONTROLS — DAFCS

67-57. ANTITORQUE CONTROLS — DAFCS.

67-58. DESCRIPTION.

The antitorque control system (figure 67-18), consists of control linkage and components from pilot and copilot pedals (20, 21 and 16, 17) aft to crosshead (1) on tail rotor. The DAFCS SAR incorporates an electromechanical servoactuator (9), rotary actuator (13), force gradient (12), and two transducers (18). The servoactuator (9) is located forward of antitorque hydraulic cylinder (7) and support. The rotary actuator (13) is located on main bulkhead under access panel below pilot seat. The force gradient (12) is connected to the rotary actuator (13) and directional control bellcrank. The two transducers (18) are connected to directional control bellcrank located aft of pilot left and right pedals (20 and 21) and to the bracket on main beam of airframe structure.

Actuation of the pilot or copilot pedals causes a hydraulic power assisted pitch change of the tail rotor blades to offset main rotor torque and to control the directional heading of the helicopter.

Dual controls are installed for copilot control of the helicopter.

Operation of antitorque dual controls system is basically the same as operation of single system. Refer to Flight Manual for operating procedures.

67-59. ANTITORQUE TRANSDUCERS.

67-60. Removal.

1. Remove access panel from cabin floor under pilot seat.

2. Disconnect electrical connectors (9 and 17, figure 67-19).

3. Remove bolt (1).

4. Remove bolt (8).

5. Remove bolt (16).

6. Remove transducers (3 and 14).

67-61. Inspection.

1. Inspect transducers (3 and 14, figure 67-19) for damage and secured electrical plug.

2. Inspect bearings for looseness (BHT-ALL-SPM).

67-62. Repair.

1. Antitorque control transducers are nonrepairable and must be replaced when defective.

67-63. Installation.

1. Position transducers (3 and 14, figure 67-19) on bracket of main beam of airframe structure.

2. Install bolt (8), aluminum washer (7), spacer (4), nut (5), and cotter pin (6).

3. Install bolt (16), aluminum washer (15), spacer (13), nut (12), and cotter pin (11).

4. Position transducers (3 and 14) on pedal adjuster (21) bellcrank.

5. Install bolt (1), aluminum washer (2), spacers (10), aluminum washer (20), nut (19), and cotter pin (18).

6. Connect electrical connectors (9 and 17).

7. Install access panel on floor under pilot seat.

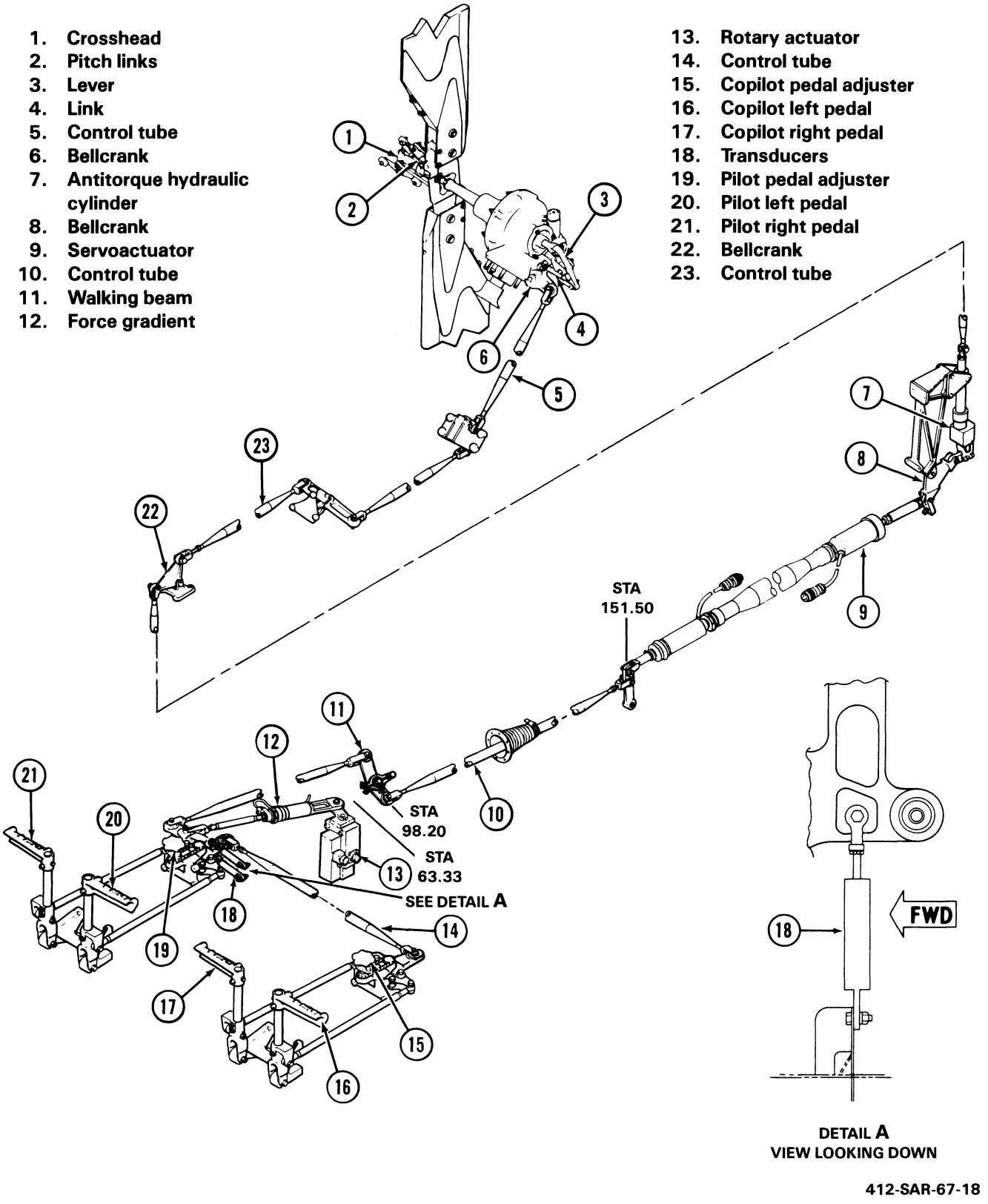
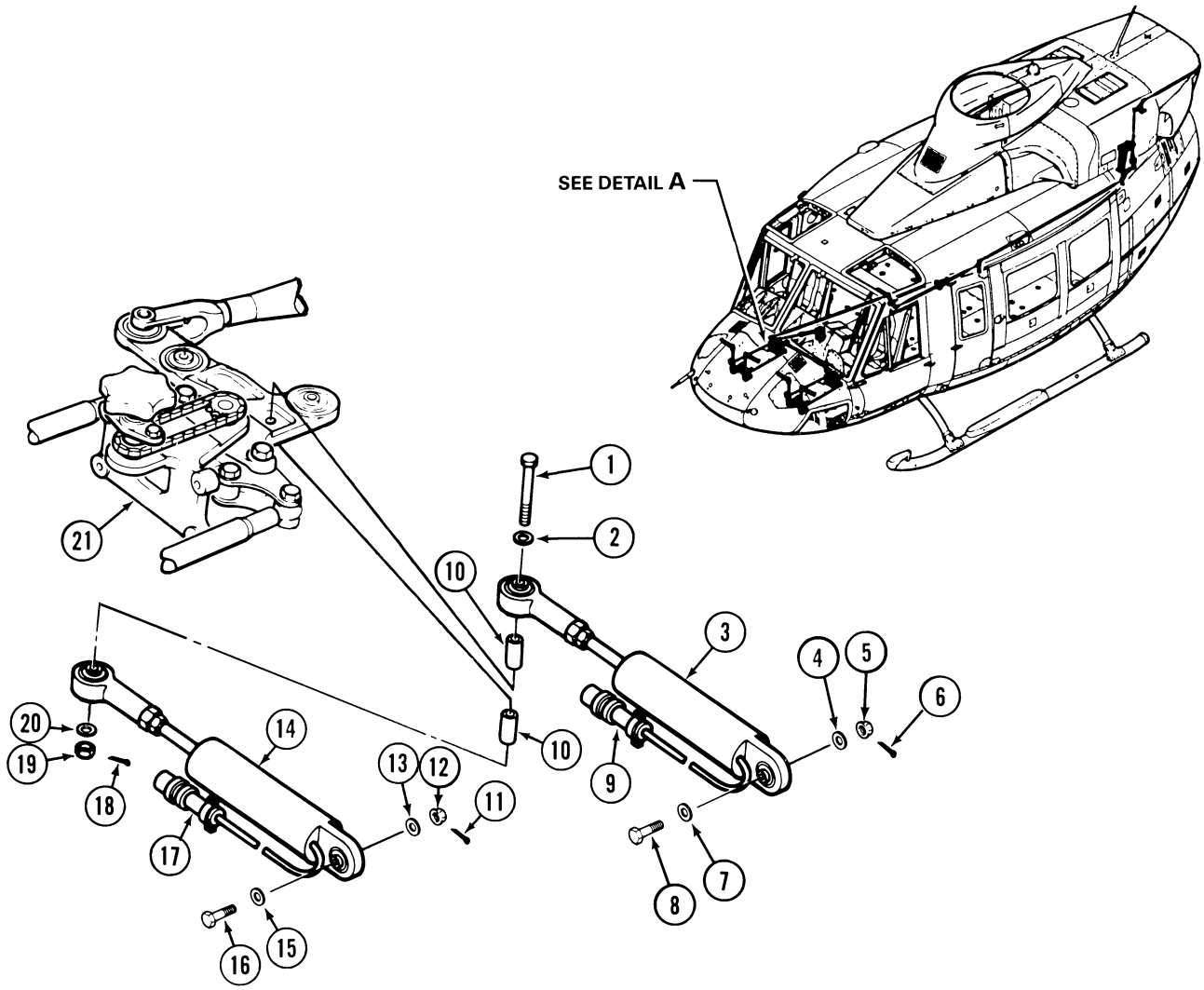


Figure 67-18. Antitorque controls



1. Bolt
2. Aluminum washer
3. Transducer
4. Spacer
5. Nut
6. Cotter pin
7. Aluminum washer
8. Bolt
9. Electrical connector
10. Spacer
11. Cotter pin
12. Nut
13. Spacer
14. Transducer
15. Aluminum washer
16. Bolt
17. Electrical connector
18. Cotter pin
19. Nut
20. Aluminum washer
21. Pedal adjuster

412-SAR-67-19

Figure 67-19. Antitorque transducer removal/installation

67-64. ANTITORQUE FORCE GRADIENT.

67-65. Removal.

1. Remove access panel from cabin floor under pilot seat.

2. Disconnect electrical connector (4, figure 67-20).

3. Remove bolt (1).

4. Remove nut (5).

5. Remove force gradient (3).

67-66. Inspection.

1. Inspect force gradient (3, figure 67-20) for corrosion and mechanical damage BHT-412-CR&O-1 . Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is not acceptable.

2. Inspect bearing for secure installation in housing and for wear (BHT-ALL-SPM).

67-67. Repair.

1. Repair of force gradient consists of replacement of unservicable components and adjustment of spring preload in accordance with paragraph 67-56 ., step 1.

67-68. Installation.

CAUTION

ENSURE THAT FORCE GRADIENT LENGTH IS ADJUSTED PRIOR TO INSTALLATION ON ROTARY ACTUATOR.

NOTE

Force gradient is similar in appearance to the force gradients used in the cyclic controls but has a different part number. Ensure that correct part number force gradient is installed for the antitorque control system.

1. If not previously accomplished, adjust spring preload on force gradient (3, figure 67-20), 18 to 20 lbs. (80.064 to 88.96 N) in accordance with paragraph 67-56 ., step 1.

2. Position force gradient (3) on pedal adjuster (12).

3. Install bolt (1), aluminum washers (2 and 9), nut (10), and cotter pin (11).

4. Position force gradient (3, figure 67-20) on stud (7) of rotary actuator (8).

5. Install nut (5), and cotter pin (6).

6. Connect electrical connector (4).

7. Move antitorque controls through full throw and check for obstructions and binding.

8. Confirm that antitorque control rigging is within limits. Refer paragraph 67-7 .

9. Install access panel on cabin floor under pilot seat.

67-69. ANTITORQUE ROTARY ACTUATOR.

67-70. Removal.

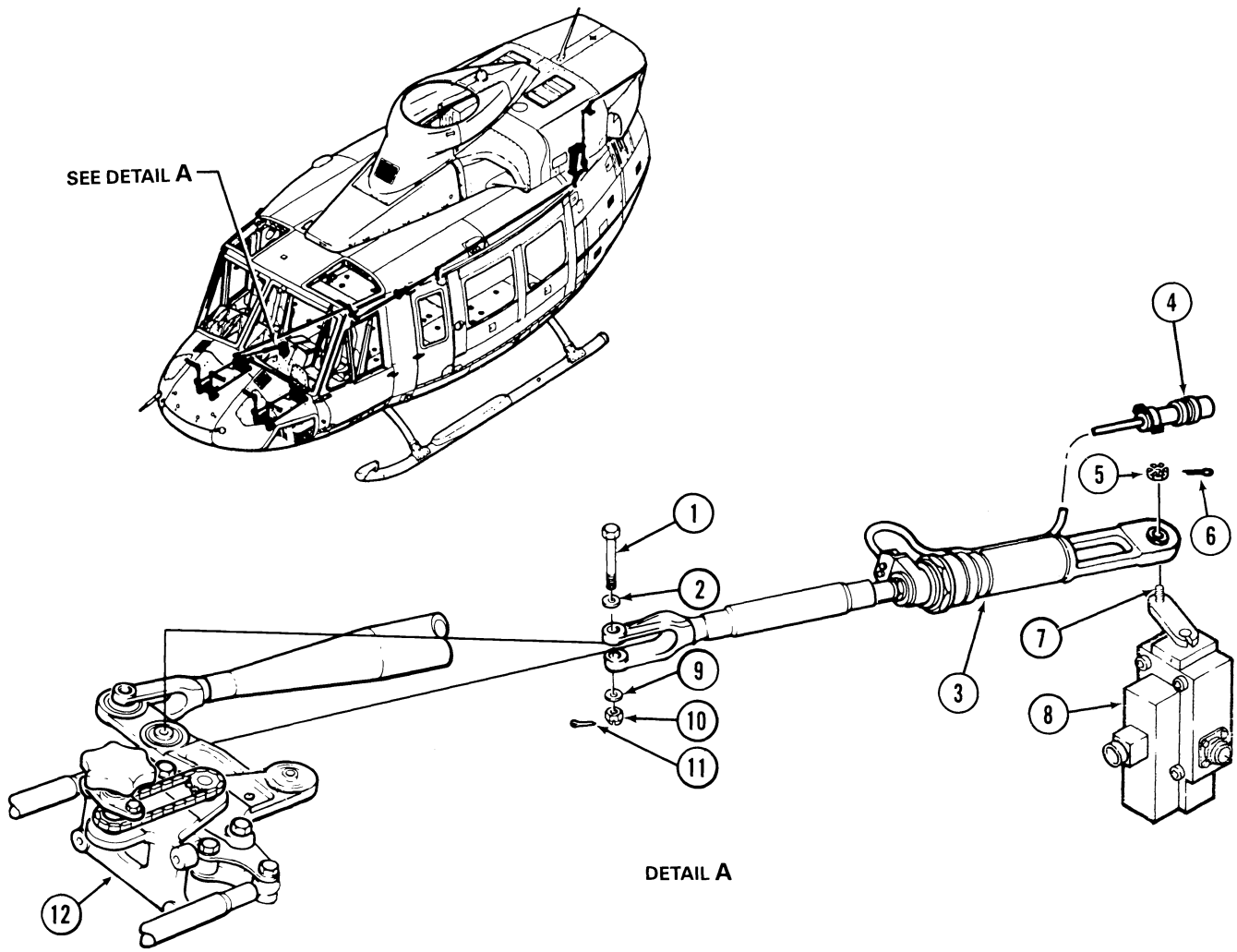
1. Remove access panel from cabin floor under pilot seat.

2. Disconnect electrical connector (6, figure 67-21).

3. Remove nut (3) from stud (4).

4. Remove four bolts (9) and four aluminum washers (8).

5. Remove rotary actuator (7) from structure (5).



1. Bolt
2. Aluminum washer
3. Force gradient
4. Electrical connector
5. Nut
6. Cotter pin
7. Stud
8. Rotary actuator
9. Aluminum washer
10. Nut
11. Cotter pin
12. Pedal adjuster

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Figure 67-20. Antitorque force gradient removal/installation

67-71. Inspection.

1. Inspect rotary actuator (7, figure 67-21) for corrosion and mechanical damage. Superficial mechanical damage is acceptable without repair. Mechanical damage in excess of superficial and/or corrosion damage is not acceptable

2. Inspect stud (4) for damaged threads. No damage that can be detected visually is acceptable.

3. Inspect electrical receptable for bent pins and threads. Superficial damage is acceptable without repair.

67-72. Repair.

Refer to vendor manual for rotary actuator repair instructions.

67-73. Installation.

1. Inspect rotary actuator (7, figure 67-21) to ensure arm (10) is properly installed on rotary actuator (7) as outlined in paragraph 67-7 rigging instructions.

2. Position rotary actuator (7) on structure (5). Install four bolts (9) and four aluminum washers (8) as follows:

3. Install aluminum washer (8) with each bolt (9).

4. Install four bolts (9) of correct length to obtain proper thread engagement. Three different length bolts are required.

5. Install bolts (9) with drilled heads in two inboard holes in rotary actuator (7) and structure (5).

6. Position force gradient (1) on stud (4) of arm (10). Install nut (3) and cotter pin (2).

7. Install electrical connector (6).

8. Move antitorque controls through full throw and check for binding and obstructions.

9. Confirm that antitorque control rigging is within limits paragraph 67-7 .

10. Install access panel on cabin floor under pilot seat.

67-74. ANTITORQUE SCAS SERVOACTUATOR.

67-75. Removal.

1. Remove access panel from bottom of aft fuselage. Open access door on right lower right side of aft fuselage.

2. Disconnect electrical connectors (8, figure 67-22).

3. Remove bolt (6).

4. Remove bolt (9).

5. Remove servoactuator (7).

67-76. Inspection.

1. Inspect servoactuator (7, figure 67-22 2) in accordance with procedures in BHT-412-CR&O-1 manual.

67-77. Installation.

1. Establish overall dimensions of servoactuator (7, figure 67-22) as shown in figure 67-23 .

2. Place servoactuator (7, figure 67-22) into position with rod end bearing in bellcrank (4).

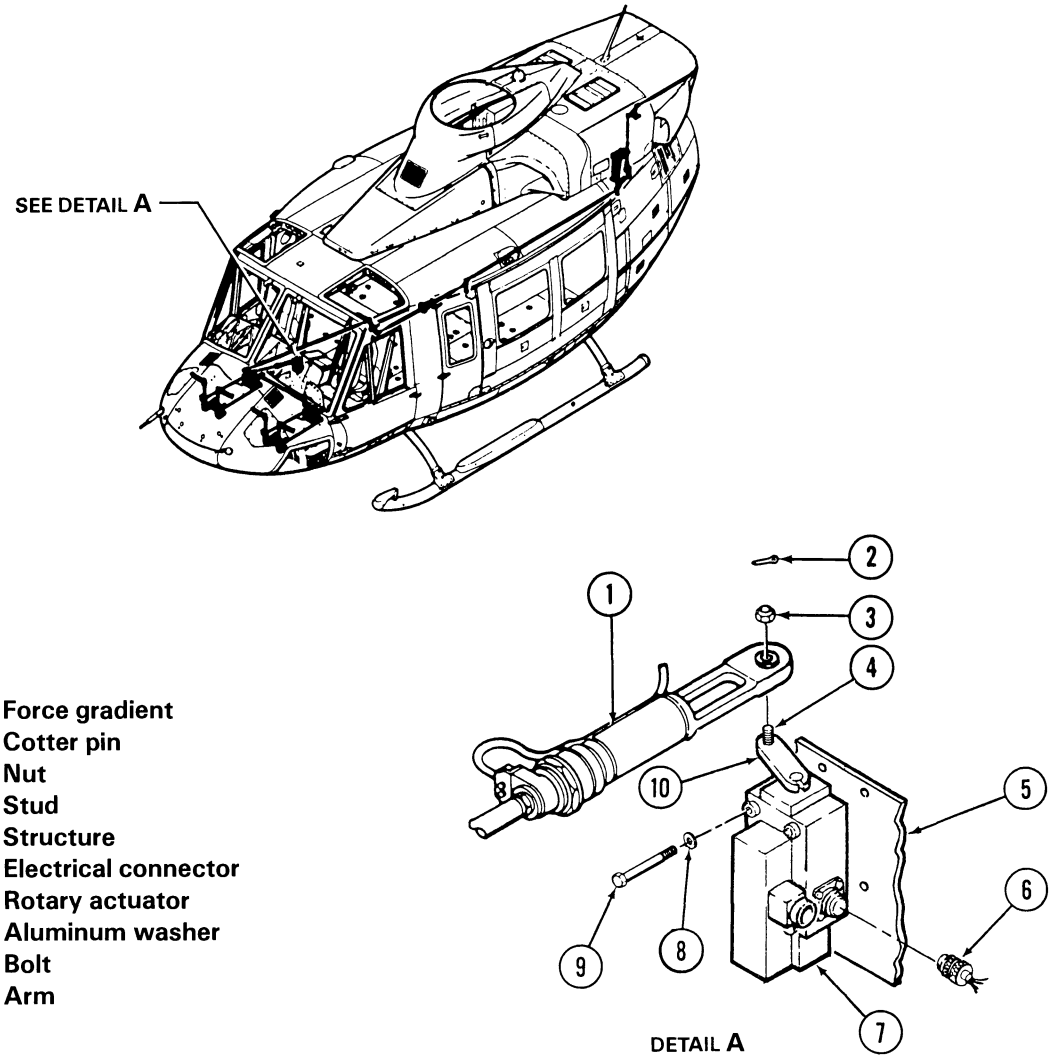
3. Install bolt (6), aluminum washers (5 and 3), nut (2), and cotter pin (1).

4. Place forward end of servoactuator (7) on lever (11).

5. Install bolt (9), aluminum washers (10 and 12), nut (13), and cotter pin (14).

6. Move antitorque controls through full travel and check for obstructions and binding.

7. Connect electrical connectors (8).



1. Force gradient
2. Cotter pin
3. Nut
4. Stud
5. Structure
6. Electrical connector
7. Rotary actuator
8. Aluminum washer
9. Bolt
10. Arm

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Figure 67-21. Antitorque rotary actuator removal/installation

8. Confirm that antitorque rigging is within limits. (Refer to paragraph 67-7).

9. Install access panel on bottom of aft fuselage.

10. Close access door on lower right side of aft fuselage.

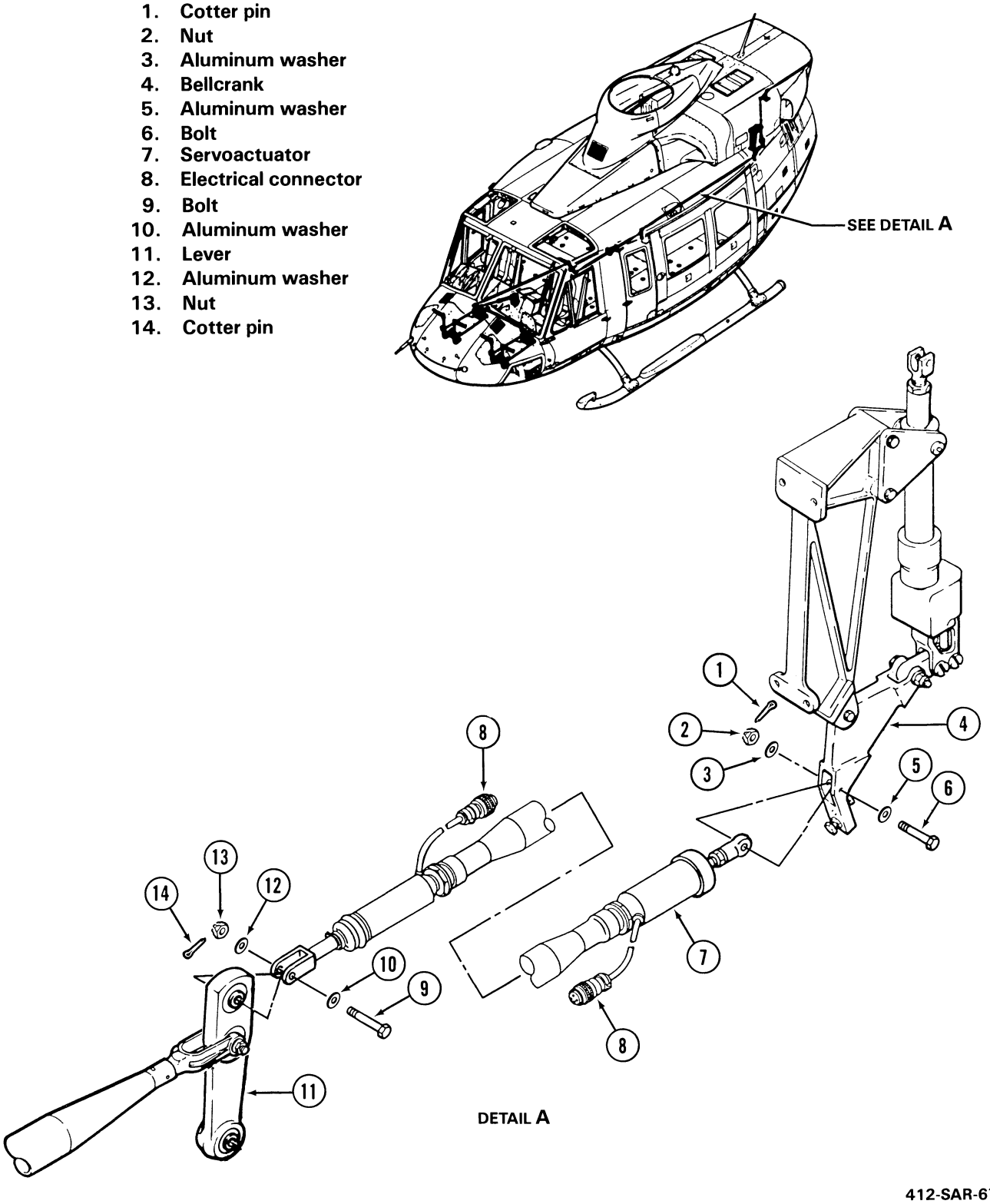
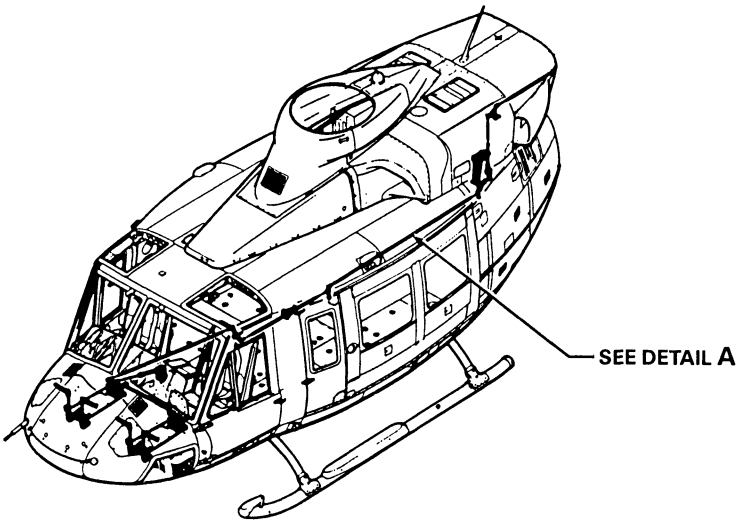
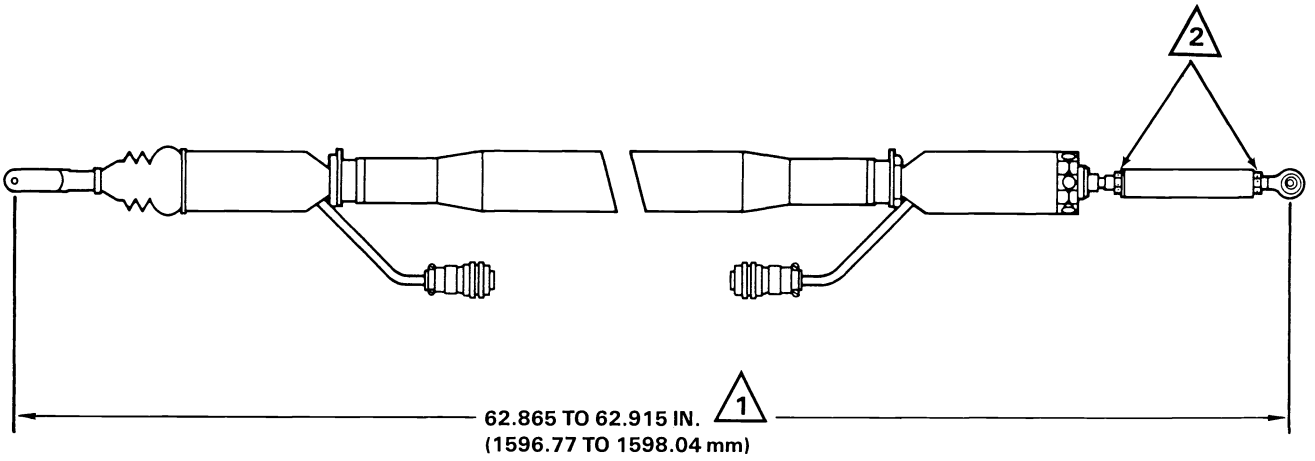


Figure 67-22. Antitorque SCAS servoactuator removal/installation





DETAIL A
412-301-700-101
DIRECTION SCAS
SERVOACTUATOR

NOTES

1. Length shown is with actuator at mid-position.
2. Nut to have full engagement of threads.
3. Actuator length is 63.259 to 63.309 inches (1606.77 to 1608.04 mm) fully extended. Minimum length is 62.471 to 62.521 inches (1586.763 to 1588.033 mm) fully retracted.

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Figure 67-23. Antitorque SCAS servoactuator